

The coppe document class

Version 3.8

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

2 License

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a mailing list where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain three-parts: *front*, *main* and *back* matters (COPPE/UFRJ Academic Registry guidelines). Each one of these parts is started by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`. The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (capa)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The capa is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation (COPPE/UFRJ Academic Registry guidelines). It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your monograph in Brazilian Portuguese and English respectively (the two built-in languages of CoppeTeX 4.x). For documents whose main language is neither

brazilian nor english – e.g. spanish, french, italian, or a user-supplied language –, also call `\titlein{<lang>}{<text>}` to register the title in that language; the cover and folha de rosto print whichever title matches the main language declared in the class option. See the section “Multilingual support” below for the complete picture. The `babel` package is loaded automatically by `coppe.cls`, so you do not need to load it yourself.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc. students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Title}{Advisor's Name}{Surname}{Degree}
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the COPPE/UFRJ Academic Registry, abstracts must be in one page each, with at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.
`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.
`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which
`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is
`\printloabbreviations` provided, it will used as sort key. This was later, in 2024, implemented also for
`\printlosymbols` `\abbrev`, otherwise `Symbol`, or `Abbreviation` will be used as sort key, what can result in an undesirable order if it contains L^AT_EX commands, mathematical symbols, or mix of uppercase and lowercase. MakeIndex needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain MakeIndex input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

COPPE_T_EX produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no Bib_T_EX `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` Bib_T_EX styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib}    % in the preamble
...
\printbibliography              % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `LATEX`, `biber`, and twice again `LATEX` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Reference entry types

Every ABNT document category maps to a `biblatex` entry type, each with an unaccented Portuguese synonym (a worked example of every type is in the sample manuscript’s reference list):

Academic `@book` (`@livro`); `@article` (`@artigo`); `@artigodejornal` (newspaper); `@incollection` (`@partedelivro`); `@inbook` (`@emlivro`); `@inreference` (`@verbete`); `@proceedings` (`@anais`); `@inproceedings` (`@trabalhodeevento`); `@thesis` (`@tese/@dissertacao`, with the presets `@dissertacaomestrado`, `@tesedoutorado`, `@teseph`, `@tcc`); `@report` (`@relatorio`); `@periodical` (`@periodico`); `@manual`; `@misc` (`@diverso`); `@online`.

Special (NBR 6023 §4.2.5–4.2.14) `@patent` (`@patente`); `@legislation` (`@legislacao/@atonormat`); `@jurisdiction` (`@jurisprudencia`, with a `relator` field); `@legal` (`@documentojuridico`); `@standard` (`@norma`); `@music` (`@partitura`); `@map` (`@mapa`); `@image` (`@iconografico`); `@artwork` (`@obradearte`); `@video` (`@audiovisual`); `@movie` (`@filme`); `@letter` (`@correspondencia`); `@dataset` (`@conjuntodedados`).

Games and TV `@game` (`@jogo`); `@boardgame` (`@jogodetabuleiro`); `@videogame` (`@jogoeletronico`); `@tvshow` (`@programadetv`).

When an entry has no `author` or `editor`, it is entered by title, with the first significant word (and a leading article) in capitals.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard `LATEX` command used before all appendices. `COPPETEX` introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

Final colophon page (`\coppetexfinalpage`)

`\coppetexfinalpage` Optional closing page modelled on the printer’s *colophon*: it records, in Portuguese, the tooling and edition used to typeset the thesis. Call it once, after every other piece of content, immediately before `\end{document}`. The macro

- forces a verso (even) page, so in a two-sided print the colophon sits opposite the rear cover;
- types the body at the foot of the page (`\vfill`);
- renders the text in Portuguese unconditionally (Brazilian `babel`), even when the thesis main language is English, Spanish, French or Italian — the colophon is institutional metadata, not body content.

Three slot macros are user-overridable. Redefine them with `\renewcommand` before `\coppetexfinalpage` if the document deviates from the defaults:

`\coppefinalengine` the T_EX engine identifier. Default: auto-detected via `iftex` (`\hologo{pdfLaTeX}`, `\hologo{LuaLaTeX}` or `\hologo{XeLaTeX}`).

`\coppefinalfont` the text body font. Default: *Computer Modern (padrão L^AT_EX)*. Redefine if you loaded `lmodern`, `mathptmx`, an OpenType math font, etc.

`\coppefinalmanual` the reference manual title (typeset in italics). Default: the current COPPE/UFRJ norm. Redefine if your work follows an earlier edition.

The macro also prints the COPPE_TE_X class version (captured into `\coppe@version` at class-load time, so it survives any later package that may redefine the global `\fileversion`) and the GitHub repository URL, <https://github.com/COPPE-UFRJ/CoppeTeX>.

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsccexam, msccexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msccexam`, and `dsccexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

brazilian, english, spanish, french, italian Pick the document’s *main* language. The class loads Babel with that language as the main one (the last in the Babel option list); inside the document `\selectlanguage{...}` and `\begin{otherlanguage}{...}` keep working as usual. The default, when no language option is given, is `brazilian`.

- `brazilian` and `english` are *built-in*: their strings ship inside `coppe.cls` and need no extra file.
- `spanish`, `french`, `italian` are shipped as *language packs*: two files per language (`coppe-lang-<lang>.def` and `<lang>-coppe.lbx`) that must sit next to `coppe.cls` (or in your project directory). They are auto-loaded by the class. Picking one of them as the main language without the matching pack files installed raises a clear class error.
- Any other Babel-known language can be plugged in by writing the same two files and either passing the language name as a class option or calling `\usecoppelanguage{<lang>}` in the preamble.

See the section “Multilingual support” below for the three-slot architecture, the `\titlein` / `\brazilianabstract` API, and how to create a new language pack.

5.1 Changing document identification

`\freeconfig` The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

5.2 Multilingual support

CoppeTeX 4.x reserves three fixed language slots:

main the language of the body, captions, TOC labels and the `abstract` environment. Picked by the class option (`brazilian` by default; one of `english`, `spanish`, `french`, `italian` also accepted out of the box; any other Babel language possible with a user-supplied pack).

foreign the language of the `foreignabstract` environment. `english` by default for every main language; switches automatically to `brazilian` when `main = english`.

third optional. Loaded on demand with `\usecoppelanguage{<lang>}` in the preamble, useful when the body quotes or cites in a third language (e.g. a Spanish-main thesis citing a French source). Does not have a class option of its own.

`\titlein`
`brazilianabstract (env.)`
`\usecoppelanguage`

Two pieces of user-facing API come with the multilingual model:

- `\titlein{<lang>}{<text>}` records the title in any language. `\title{...}` still writes the Brazilian-Portuguese title and `\foreigntitle{...}` still writes the English one (back-compat); for a main language outside the `brazilian/english` pair, also call `\titlein{<lang>}{...}`. The cover and folha de rosto print whichever title matches `\coppe@mainlang`.
- The `brazilianabstract` environment is the optional third abstract used when the main language is neither `brazilian` nor `english` – typically a Spanish/French/Italian thesis that also wants a Portuguese resumo for UFRJ examiners. It wraps its body in `\begin{otherlanguage}{brazilian}` and uses the Portuguese template strings. A document with `main = brazilian` can simply keep using `abstract`; `brazilianabstract` is redundant then.
- `\usecoppelanguage{<lang>}` loads an extra language pack in the preamble without making it the main or foreign language.

Institutional names stay Portuguese

The cover, folha de rosto and ficha catalográfica are a Brazilian institutional template (NBR 14724 + COPPE manual). The strings `universityname`, `cityname`, `statename`, `countryname`, the departments declared with `\department` and the surrounding “apresentada/submetida ao...” wording are deliberately kept in Portuguese for every main language. Only the title (printed via `\coppe@selecttitle`) follows the main language.

Writing a new language pack

A language pack is just two files dropped next to `coppe.cls` (or anywhere on the `TEXINPUTS` path):

`coppe-lang-<lang>.def` class-level strings. Populate the dispatcher table with `\copperdefstring{<lang>}{<key>}{<value>}` for every key already populated for `brazilian/english` in `coppe.cls` (appendix, annex, frame, listframe, source, program, listprogram, references, in:, art, platform, directedby, producedby, listabbreviation, listsymbol, glossary, advisor, advisors, dept, approvedmale, approvedfemale, universityname, cityname, statename, countryname, monthname1...monthname12, algorithm2eopt, babelname). Finish with

```
\AtBeginDocument{\DeclareLanguageMapping{<lang>}{<lang>-coppe}}
```

so the `biblatex` mapping is registered after the bibliography engine is loaded by `coppe.cls` (which happens *after* the language pack is read).

`<lang>-coppe.lbx` `biblatex` localization strings. `\InheritBibliographyExtras{<lang>}` then `\DeclareBibliographyStrings` for the keys `availablefrom`, `urlseen`, `in`, `editor`, `editors`, `depositor`, `coppescale`, `mscdiss`, `dscthesi`, `phdthesis`, `tcc`. The shipped `spanish-coppe.lbx`, `french-coppe.lbx` and `italian-coppe.lbx` are the reference templates to copy from.

Tip: the `algorithm2eopt` key holds whatever `algorithm2e` calls the language. Mind the spelling: `algorithm2e` ships `portuguese`, `english`, `spanish`, `french` and `italiano` – not `italian`.

Example: a Spanish-main thesis

```
\documentclass[spanish,dsc]{coppe}
\addbibresource{example.bib}
\begin{document}
  \title{T'\i tulo em portugu\~es}      % Brazilian abstract slot
  \foreigntitle{English title}          % English abstract slot
  \titlein{spanish}{T'\i tulo en espa\~nol} % cover/folha
  \author{Juan}{P'\erez}
  \advisor{Prof.}{Mar'\i a}{Garc'\i a}{D.Sc.}
  \department{PESC}\date{01}{2026}
  \keyword{multilenguaje}
  \maketitle
  \frontmatter
  \begin{abstract}      Resumen principal en espa\~nol.    \end{abstract}
  \begin{foreignabstract} English foreign abstract.        \end{foreignabstract}
  \begin{brazilianabstract} Resumo em portugu\~es para a banca. \end{brazilianabstract}
  \tableofcontents
  \mainmatter
  \chapter{Introducci\on} Cuerpo en espa\~nol\dots
\end{document}
```

6 Files in your project folder

In a normal install where CoppeTeX is part of your `TEXINPUTS` path (e.g. inside the local `texmf` tree of MiKTeX or TeX Live), you do not need to think about this list at all — the files travel with the class. This section is for two situations where you do:

- shipping the class together with your manuscript (a self-contained submission package); or
- installing CoppeTeX by hand, dropping the files next to `coppe.cls` in the project directory.

Always required

These files must be findable at every compilation, regardless of the main language or any class option:

`coppe.cls` the document class itself.

`coppe.dbx` custom `biblatex` datamodel (declares the ABNT-specific entry types `@game`, `@videogame`, `@boardgame`, `@tvshow` and `@map`, plus a few custom fields).

`coppe.bbx` and `coppe.cbx` `biblatex` bibliography and citation styles for the author–date scheme (the default).

`brazilian-coppe.lbx` and `english-coppe.lbx` `biblatex` localisation tables for Portuguese and English. Both are required *even when neither language is the main one*, because `coppe.bbx` registers

```
\DeclareLanguageMapping{brazilian}{brazilian-coppe}
\DeclareLanguageMapping{english}{english-coppe}
```

unconditionally, so `biblatex` tries to read both LBX files at every compilation.

Conditional on class options

`coppe-numeric.bbx` and `coppe-numeric.cbx` only when the `numbers` class option is given (`\documentclass[numbers]{coppe}`). Without it, `biblatex` never tries to read them.

`coppe.ist` only when the glossary, list-of-symbols, or list-of-abbreviations features are used. It is the `makeindex` style file invoked by the `makeindex -s coppe.ist ...` step described in subsection *Lists of symbols and abbreviations* above.

Conditional on the main language

The main language is chosen by the matching class option (`brazilian`, `english`, `spanish`, `french`, `italian`). The two built-in languages ship their strings inside `coppe.cls` and so need no extra file. The three language packs add two files each — one class-level (`coppe-lang-<lang>.def`) and one `biblatex`-level (`<lang>-coppe.lbx`):

Main language	Extra files needed
<code>brazilian</code> (default)	— (built into <code>coppe.cls</code>)
<code>english</code>	— (built into <code>coppe.cls</code>)
<code>spanish</code>	<code>coppe-lang-spanish.def</code> , <code>spanish-coppe.lbx</code>
<code>french</code>	<code>coppe-lang-french.def</code> , <code>french-coppe.lbx</code>
<code>italian</code>	<code>coppe-lang-italian.def</code> , <code>italian-coppe.lbx</code>

Loading a third language at preamble time via `\usecoppelanguage{<lang>}` brings in the same pair of files for that language. A user-supplied language pack follows the same scheme: drop `coppe-lang-<lang>.def` and `<lang>-coppe.lbx` next to `coppe.cls`.

Cover assets

The institutional title page produced by `\maketitle` needs two PDF logo files at compile time:

`ufrj-logo.pdf` and `coppe-logo.pdf` the UFRJ shield and the COPPE/UFRJ wordmark. Both ship with the CoppeTeX distribution; they must be findable from the project directory (or on `TEXINPUTS`).

7 Quick, useful tips

Pictures. The default picture format of $\text{\LaTeX}$ is the Encapsulated PostScript (EPS). If you use `pdf $\text{\LaTeX}$` , the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and `pdflatex` will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in $\text{\LaTeX}$ is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package, which is loaded by the `coppe` class itself (since v4.0): the user must *not* load it again in the manuscript preamble. Author, title, subject and keyword metadata are filled in automatically by `\maketitle`.

Math (`amssymb` vs. `unicode-math`). The class always loads `amsmath`. The companion AMS symbol package `amssymb`, however, is loaded *only* under `pdf $\text{\LaTeX}$` : it is a legacy 8-bit package built around the `msam/msbm` Type 1 fonts and clashes with `unicode-math`, the modern OpenType-math package used on Unicode engines (Lua $\text{\LaTeX}$, Xe $\text{\LaTeX}$). On those engines the class skips `amssymb` and schedules an *end-of-document* check: if by then the author has loaded neither `unicode-math` nor `amssymb` (which they could have loaded by hand, though it is discouraged), a `ClassWarning` reminds them to add

```
\usepackage{unicode-math}
\setmathfont{Latin Modern Math} % or STIX Two Math, ...
```

to the manuscript preamble. The class deliberately does *not* auto-load `unicode-math`, because picking an OpenType math font is a typographic choice that belongs to the author. **Never load `amssymb` together with `unicode-math`:** it would re-establish the 8-bit AMS fonts on top of the OpenType math font and split symbols across two incompatible encodings.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

longquote Quotation To quote text larger than three lines, according to ABNT, you must increase the left margin to 4 cm, do not use quotation marks, and use a smaller font. The `coppe` class provides the `longquote` environment to easily make these adjustments.

8 A simple example

```

1 \example
2 \documentclass[dsc]{coppe}
3
4 \usepackage{rotating}          % rodando coisas, como tabelas
5 \usepackage[most]{tcolorbox} % caixas de texto
6 \usepackage{tikz}             % gráficos vetoriais (o tcolorbox já o traz)
7 % Sob LuaLaTeX/XeLaTeX, descomente as duas linhas a seguir para obter os
8 % símbolos AMS (\mathbb, \hbar, \varnothing, ...) via unicode-math. Sob
9 % pdfLaTeX, a classe já carrega amssymb automaticamente; deixe comentado.
10 %\usepackage{unicode-math}
11 %\setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
12 \DeclareMathOperator{\RMSE}{RMSE} % operadores próprios (cap. de matemática)
13 \DeclareMathOperator{\diag}{diag}
14 % A classe coppe (v4.0) já carrega: amsmath, booktabs, longtable, subcaption,
15 % hyperref, biblatex, listings, fancyvrb, algorithm2e -- não recarregue esses
16 % pacotes aqui (em particular, hyperref deve ficar a cargo da classe para que
17 % a ordem com setspace e biblatex permaneça correta).
18
19 \addbibresource{example.bib}% base de referências (biblatex/biber)
20
21 \makelossymbols
22 \makeloabbreviations
23
24 \begin{document}
25   \title{Título da Tese}
26   \foreigntitle{Thesis Title}
27   \author{Nome do Autor}{Sobrenome}
28   \advisor{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
29   \advisor{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
30   % A classe aceita de um a tres \advisor; se houver apenas um, o rotulo na
31   % capa e na folha de rosto sai como "Orientador" (singular), com dois ou
32   % mais sai como "Orientadores" (plural). Para o caso 1/3 orientadores,
33   % veja tests/test_brazilian_one_advisor.tex e tests/test_brazilian_three_advisors.tex.
34
35   \examiner{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
36   \examiner{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
37   \examiner{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
38   \examiner{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
39   \examiner{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
40   \department{PESC}
41   \date{01}{2024}

```

```

42
43 \keyword{Primeira palavra-chave}
44 \keyword{Segunda palavra-chave}
45 \keyword{Terceira palavra-chave}
46
47 \maketitle
48
49 \frontmatter
50 \dedication{A alguém cujo valor é digno desta dedicatória.}
51
52 \chapter*{Agradecimentos}
53
54 Gostaria de agradecer a todos os que contribu\iram para este trabalho:
55 orientadores, colegas, familiares e \as ag\ências de fomento. Lorem ipsum
56 dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
57 ut labore et dolore magna aliqua.
58
59 \begin{abstract}
60
61 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
62 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
63 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
64 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
65 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
66 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
67 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
68 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
69 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
70 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
71 aspernatur aut odit aut fugit.
72
73 \end{abstract}
74
75 \begin{foreignabstract}
76
77 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
78 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
79 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
80 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
81 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
82 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
83 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
84 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
85 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
86 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
87 aspernatur aut odit aut fugit.
88
89 \end{foreignabstract}
90
91 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
92 % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
93 \listoffigures
94 \listoftables
95 \listofquadros

```

```

96 \listofprogramas
97 \listofalgorithms
98 \printloabbreviations
99 \printlosymbols
100 \tableofcontents
101
102 \mainmatter
103 \chapter{Introdução}
104
105 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos da
106
107 A classe \verb|coppe| foi criada por Vicente Helano e George Ainsworth, porém, em 2024, é
108
109 A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.com}
110
111 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po
112
113 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p
114
115 É importante de notar que essa classe não foi construída sobre a classe \LaTeX \ para a A
116
117 Apesar desse modelo ser muito bom, ele tem um defeito: a limitação do sistema de referênci
118
119 Mais ainda, as regras da COPPE ainda não se adaptaram, no início de 2024, as novas regras
120
121 Este documento não substitui, mas complementa, o documento que descreve a classe.
122
123
124
125 \chapter{Configurações Iniciais}
126
127 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c
128
129 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra
130
131 Recomendo ler o documento “The \verb*|coppe| document class” para entender melhor todas
132
133 \section{Linguagem principal do texto}
134
135 Essa classe considera que o texto principal está em português e algumas partes específicas
136 \begin{itemize}
137 \item A opção \verb|english| deve ser usada no comando \verb|\documentclass|.
138 \item A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p
139 \end{itemize}
140
141 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe \CoppeT
142
143
144 \section{Por que usar o \LaTeX}
145
146 Há uma grande discussão entre usuário de Word e \LaTeX, principalmente, quanto ao uso dess
147
148 Nós escolhemos o \LaTeX por alguns motivos: grande facilidade de seguir um estilo sem se p
149

```

150 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob
151
152 \section{Como e onde usar o \LaTeX}
153
154 Existem muitos tutoriais de \LaTeX, mas basicamente, em 2024, ele é usado em dois ambientes
155 \begin{enumerate}
156 \item Na sua máquina, instalando uma versão completa como o \hologo{MiKTeX}, típico do Windows
157 \item Usar um ambiente na rede, como o Overleaf.
158 \end{enumerate}
159
160 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o backup
161
162
163 \chapter{Algumas Regras da COPPE}
164
165 Todas abreviaturas e símbolos devem ser definidas antes de utilizadas. Isso é facilmente feito
166
167 É imprescindível definir os símbolos, tal como o
168 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.
169 \syml{\mathbb{R}}{Conjunto dos números reais}
170 \syml{\emptyset}{Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como
171
172 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o \verb|makeindex|
173
174 Como as listas de símbolos e de abreviaturas usamos o mesmo comando usado para criar índices
175
176 \section{Citações}
177 \label{sec:citacoes}
178
179 Citações curtas podem ser feitas \code{\quote{...}} ou direto com “duas crases e dois pontos”
180
181 \begin{longquote}
182 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
183 feita com o ambiente \code{\verb=longquote=}: The primary objective of this
184 investigation was to determine the feasibility of detecting corrosion in
185 aluminum Naval aircraft components with neutron radiographic interrogation
186 and the use of standard corrosion penetrameters. Secondary objectives
187 included the determination of the effect of object thickness on image quality,
188 the defining of minimum levels of detectability and a preliminary investigation
189 of a means whereby the degree of corrosion could be quantified with neutron
190 radiographic data. \code{\cite{article-example}}
191 \end{longquote}
192
193 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote \code{\verb*|natbib|}
194
195 Em todo caso, \code{\textbf{deve}} se tomar enorme atenção com as citações, para evitar ocorrer erros
196
197 Quando se cita uma obra por meio de outra (\code{\textit{apud}}), pela ABNT entra na lista de referências
198
199 \chapter{Floats}
200
201 Grande parte dos problemas de iniciantes, e veteranos, em \LaTeX é da localização dos \code{\textit{floats}}
202
203 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesma

```

204
205 \textbf{Pela norma atual (manual2024), a legenda} \verb|\caption| \textbf{de toda ilustração}
206
207 Quadros são ilustrações cujos dados são delimitados por linhas em \emph{todas} as bordas (ao
208
209
210 \section{Tabelas e Figuras Padrão}
211
212 Vamos ver uma tabela padrão, como a \autoref{tab:exemplo_numeros}.
213
214 \begin{table}[ht]
215 \centering % Centraliza a tabela
216 \caption{Exemplo de Tabela de Números}
217 \label{tab:exemplo_numeros}
218 \begin{tabular}{ccc} % Define a quantidade de colunas
219 \hline % Linha superior
220 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ % Cabeçalhos
221 \hline % Linha média
222 1 & 2 & 3 \\ % Primeira linha de dados
223 \hline
224 4 & 5 & 6 \\ % Segunda linha de dados
225 \hline
226 7 & 8 & 9 \\ % Terceira linha de dados
227 \hline
228 10 & 11 & 12 \\ % Quarta linha de dados
229 \hline % Linha inferior
230 \end{tabular}
231 \fonte{Elabora\c{c}\~ao pr\'opria.} % fonte obrigatoria, abaixo do float
232 \end{table}
233
234
235
236 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
237
238 \begin{figure}[ht]
239 \centering % Centraliza a figura
240 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
241 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
242 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
243 \source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura
244 \end{figure}
245
246 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
247
248 \begin{quadro}[ht]
249 \centering
250 \caption{Exemplo de Quadro}
251 \label{qua:exemplo}
252 \begin{tabular}{|l|l|}
253 \hline
254 \textbf{Ilustração} & \textbf{Delimitação} \\
255 \hline
256 Tabela & Linhas apenas no topo e na base. \\
257 \hline

```

```

258 Quadro & Linhas em todas as bordas. \\
259 \hline
260 \end{tabular}
261 \fonte{Elabora\c{c}\~ao pr\'opria.}
262 \end{quadro}
263
264
265
266
267
268 \section{Tabelas mais elegantes}
269
270 Atualmente a tend\^encia \e usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
271
272 \begin{table}[ht]
273 \centering % Centraliza a tabela
274 \caption{Exemplo de Tabela de N\^umeros mais elegantes}
275 \label{tab:exemplo_numerosbom}
276 \begin{tabular}{ccc} % Define a quantidade de colunas
277 \toprule % Linha superior
278 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \textbf{Cabe\c{c}alhos}
279 \midrule % Linha m\^edia
280 1 & 2 & 3 \\ % Primeira linha de dados
281 4 & 5 & 6 \\ % Segunda linha de dados
282 7 & 8 & 9 \\ % Terceira linha de dados
283 10 & 11 & 12 \\ % Quarta linha de dados
284 \bottomrule % Linha inferior
285 \end{tabular}
286 \fonte{Elabora\c{c}\~ao pr\'opria.}
287 \end{table}
288
289 \section{Tabelas Longas ou Largas}
290
291 Se sua tabela \e muito longa ou larga, existem v\^arias op\c{c}o\~es.
292 \begin{itemize}
293 \item alterar o tamanho da letra
294 \item Usar o longtable
295 \item rodar a tabela, fazendo ela em \textit{landscape}
296 \item fazer a tabela dentro de um minibox
297 \end{itemize}
298
299
300 \subsection{Tabelas largas demais}
301
302 \E comum em teses que as tabelas sejam largas demais. H\^a v\^arias formas de resolver isso.
303
304 A \autoref{tab:tabela_largafns} \e larga demais, e nela isso \e resolvido diminuindo a fonte p
305
306 \begin{table}[ht]
307 \centering % Centraliza a tabela
308 \caption{Exemplo de Tabela Larga com Fonte Menor}
309 \label{tab:tabela_largafns}
310 \footnotesize % Aplica uma fonte menor para a tabela
311 \begin{tabular}{ccccccc} % Aumente o n\^umero de colunas conforme necess\^ario

```

```

312 \toprule
313 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} & \textbf{Coluna 4} & \textbf{Coluna 5} \\
314 \midrule
315 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
316 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
317 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
318 \bottomrule
319 \end{tabular}
320 \fonte{Elabora\c{c}\~ao pr\'opria.}
321 \end{table}
322
323 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer conteúdo.
324
325 \begin{table}[ht]
326 \centering
327 \caption{Exemplo de Tabela Redimensionada}
328 \label{tab:examplerb}
329 \resizebox{\textwidth}{!}{%
330 \begin{tabular}{lllll}
331 \toprule
332 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 & Coluna 5 \\
333 \midrule
334 Dados 1 & Dados 2 & Dados 3 & Dados 4 & Dados 5 \\
335 Dados 6 & Dados 7 & Dados 8 & Dados 9 & Dados 10 \\
336 \bottomrule
337 \end{tabular}}%
338 }
339 \fonte{Elabora\c{c}\~ao pr\'opria.}
340 \end{table}
341
342
343 Para rodar uma tabela muito larga em 90 graus no LaTeX, você pode usar o pacote \verb*|rotatble|.
344
345 Aqui está um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela.
346
347 \begin{sidewaystable}
348 \centering
349 \caption{Sua Legenda Aqui}
350 \label{tab:sua_tabela}
351 \begin{tabular}{llll}
352 \toprule
353 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
354 \midrule
355 Item 1 & Item 2 & Item 3 & Item 4 \\
356 Item 5 & Item 6 & Item 7 & Item 8 \\
357 \bottomrule
358 \end{tabular}
359 \fonte{Elabora\c{c}\~ao pr\'opria.}
360 \end{sidewaystable}
361
362 Se a tabela for muito longa, o ambiente \verb|longtable| é o ideal. Ele fornece comandos para
363
364 % Exemplo de tabela longa que se estende por várias páginas
365 \begin{longtable}{|c|c|c|}

```

```

366 % primeiro cabeçalho (é o caption)
367 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
368 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
369 \endfirsthead
370 % cabeçalho normal
371 \multicolumn{3}{c}%
372 {\table\thetable} -- continuação da página anterior} \\
373 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
374 \endhead
375 % pé normal
376 \hline \multicolumn{3}{r}{{Continua na próxima página}} \\ \hline
377 \endfoot
378 \hline
379 % último pé
380 \multicolumn{3}{r}{{Continua na próxima página}} \\
381 \hline \hline
382 \endlastfoot
383
384 % Conteúdo da tabela
385 1 & 2 & 3 \\
386 4 & 5 & 6 \\
387 1 & 2 & 3 \\
388 4 & 5 & 6 \\
389 1 & 2 & 3 \\
390 4 & 5 & 6 \\
391 1 & 2 & 3 \\
392 4 & 5 & 6 \\
393 1 & 2 & 3 \\
394 4 & 5 & 6 \\
395 1 & 2 & 3 \\
396 4 & 5 & 6 \\
397 1 & 2 & 3 \\
398 4 & 5 & 6 \\
399 1 & 2 & 3 \\
400 4 & 5 & 6 \\
401 1 & 2 & 3 \\
402 1 & 2 & 3 \\
403 4 & 5 & 6 \\
404 1 & 2 & 3 \\
405 4 & 5 & 6 \\
406 1 & 2 & 3 \\
407 4 & 5 & 6 \\
408 1 & 2 & 3 \\
409 4 & 5 & 6 \\
410 1 & 2 & 3 \\
411 4 & 5 & 6 \\
412 1 & 2 & 3 \\
413 4 & 5 & 6 \\
414 1 & 2 & 3 \\
415 4 & 5 & 6 \\
416 1 & 2 & 3 \\
417 4 & 5 & 6 \\
418 1 & 2 & 3 \\
419 4 & 5 & 6

```

420 1 & 2 & 3 \\
 421 4 & 5 & 6 \\
 422 1 & 2 & 3 \\
 423 4 & 5 & 6 \\
 424 1 & 2 & 3 \\
 425 4 & 5 & 6 \\1 & 2 & 3 \\
 426 4 & 5 & 6 \\
 427 1 & 2 & 3 \\
 428 4 & 5 & 6 \\
 429 1 & 2 & 3 \\
 430 1 & 2 & 3 \\
 431 4 & 5 & 6 \\
 432 1 & 2 & 3 \\
 433 4 & 5 & 6 \\
 434 1 & 2 & 3 \\
 435 4 & 5 & 6 \\
 436 1 & 2 & 3 \\
 437 4 & 5 & 6 \\
 438 1 & 2 & 3 \\
 439 4 & 5 & 6 \\
 440 1 & 2 & 3 \\
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 451 4 & 5 & 6 \\
 452 1 & 2 & 3 \\
 453 4 & 5 & 6 \\
 454 1 & 2 & 3 \\
 455 4 & 5 & 6 \\
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 457 4 & 5 & 6 \\
 458 1 & 2 & 3 \\
 459 4 & 5 & 6 \\
 460 1 & 2 & 3 \\
 461 4 & 5 & 6 \\
 462 1 & 2 & 3 \\
 463 4 & 5 & 6 \\
 464 1 & 2 & 3 \\
 465 4 & 5 & 6 \\
 466 1 & 2 & 3 \\
 467 4 & 5 & 6 \\
 468 1 & 2 & 3 \\
 469 4 & 5 & 6 \\
 470 1 & 2 & 3 \\
 471 4 & 5 & 6 \\
 472 1 & 2 & 3 \\
 473 4 & 5 & 6 \\

```

474 1 & 2 & 3 \\
475 4 & 5 & 6 \\
476 1 & 2 & 3 \\
477 4 & 5 & 6 \\
478 1 & 2 & 3 \\
479 4 & 5 & 6 \\
480 1 & 2 & 3 \\
481 4 & 5 & 6 \\
482 1 & 2 & 3 \\
483 4 & 5 & 6 \\
484 1 & 2 & 3 \\
485 4 & 5 & 6 \\
486 1 & 2 & 3 \\
487 4 & 5 & 6 \\
488 1 & 2 & 3 \\
489 4 & 5 & 6 \\
490 1 & 2 & 3 \\
491 4 & 5 & 6 \\
492
493 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
494 \end{longtable}
495 \fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente
496
497 \section{Subfiguras (subcaption)}
498
499 Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use
500
501 \begin{figure}[ht]
502   \centering
503   \begin{subfigure}[b]{0.42\textwidth}
504     \centering\includegraphics[width=\linewidth]{example-image-a}
505     \caption{Primeira vista}\label{fig:sub-a}
506   \end{subfigure}
507   \hfill
508   \begin{subfigure}[b]{0.42\textwidth}
509     \centering\includegraphics[width=\linewidth]{example-image-b}
510     \caption{Segunda vista}\label{fig:sub-b}
511   \end{subfigure}
512   \caption{Figura com duas subfiguras}
513   \label{fig:subs}
514   \fonte{Elabora\c{c}\~ao pr\'opria.}
515 \end{figure}
516
517 Referencie a figura inteira com \verb|\autoref{fig:subs}| e cada parte com \verb|\autoref{fig:sub-a}|
518
519 \section{Imagens (graphicx)}
520
521 A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
522 \begin{itemize}
523   \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;
524   \item \verb|[angle=90]| --- rotação;
525   \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, direita, topo});
526 \end{itemize}
527 Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \texttt{eps}

```

```

528
529 \section{Introdução ao tikz}
530
531 0 \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no próprio
532
533 \begin{figure}[ht]
534   \centering
535   \begin{tikzpicture}
536     \node[draw, rounded corners] (a) at (0,0) {Início};
537     \node[draw, rounded corners] (b) at (4,0) {Fim};
538     \draw[->, thick] (a) -- (b) node[midway, above] {processo};
539   \end{tikzpicture}
540   \caption{Diagrama simples em tikz}
541   \label{fig:tikz}
542   \fonte{Elabora\c{c}\~ao pr\'opria.}
543 \end{figure}
544
545 Para gráficos de dados (curvas, barras), veja o \verb|pgfplots|, construído sobre o \verb|tikz|
546
547 \section{Float ou direto no texto?}
548
549 Colocar a tabela ou imagem dentro de \verb|figure|/\verb|table| (um \emph{float}) deixa o \verb|L|
550
551 \chapter{Using BibLaTeX}
552 \label{cap:biblatex}
553
554 Esta classe gera as referências com BibLaTeX e o \emph{backend} \texttt{biber}, no padrão
555
556 \textbf{É} obrigatório compilar com \texttt{biber} (e não com \hologo{BibTeX}). A sequência
557
558 Os comandos do \texttt{natbib} (\verb|\citet|, \verb|\citep|) continuam disponíveis porque
559
560 \begin{table}[h]
561 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao}
562   \texttt{\textbackslash cite} do \LaTeX\ e
563   o comando \texttt{\textbackslash citet},
564   disponível pela opção \texttt{natbib} do BibLaTeX.
565 \label{tab:citation}
566 \centering
567 {\footnotesize
568 \begin{tabular}{|c|c|c|}
569   \hline
570   Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
571   \hline
572   Livro & \cite{book-example} & \citet{book-example}\\
573   Artigo & \cite{article-example} & \citet{article-example}\\
574   Relatório & \cite{techreport-example} & \citet{techreport-example}\\
575   Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
576   Anais de Congresso & \cite{inproceedings-example} & \\
577   & \citet{inproceedings-example}\\
578   Séries & \cite{incollection-example} & \citet{incollection-example}\\
579   Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
580   Dissertação de mestrado & \cite{mastersthesis-example} & \\
581   & \citet{mastersthesis-example}

```

```

582     Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
583     \hline
584 \end{tabular}}
585 \fonte{Elabora\c{c}\~ao pr\'opria.}
586 \end{table}
587
588 \begin{table}[h]
589 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao
590 \texttt{\textbackslash cite} do \LaTeX\ e
591 o comando \texttt{\textbackslash citet},
592 disponivel pela op\c{c}\~ao \texttt{\natbib} do BibLaTeX. Al\~em disso, usando o booktabs.}
593 \label{tab:citation1}
594 \centering
595 {\footnotesize
596 \begin{tabular}{ccc}
597 \toprule
598 Tipo da Publica\c{c}\~ao & \verb|\cite| & \verb|\citet|\\
599 \midrule
600 Livro & \cite{book-example} & \citet{book-example}\\
601 Artigo & \cite{article-example} & \citet{article-example}\\
602 Relat\~orio & \cite{techreport-example} & \citet{techreport-example}\\
603 Relat\~orio & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
604 Anais de Congresso & \cite{inproceedings-example} &
605 \citet{inproceedings-example}\\
606 S\~eries & \cite{incollection-example} & \citet{incollection-example}\\
607 Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
608 Disserta\c{c}\~ao de mestrado & \cite{mastersthesis-example} &
609 \citet{mastersthesis-example}\\
610 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
611 \bottomrule
612 \end{tabular}}
613 \fonte{Elabora\c{c}\~ao pr\'opria.}
614 \end{table}
615
616 \section{Comandos de cita\c{c}\~ao}
617
618 Os principais comandos s\~ao: \verb|\cite| (refer\~encia simples), \verb|\citet| (autor por ex
619
620 \subsection{Par\~ametro opcional: p\~agina e ‘‘tradu\c{c}\~ao nossa’’}
621
622 Todos os comandos aceitam um par\~ametro opcional para a localiza\c{c}\~ao (p\~agina, se\c{c}\~ao) e obser
623
624 \subsection{Cita\c{c}\~ao de cita\c{c}\~ao (\emph{apud})}
625
626 Para citar uma obra atrav\~es de outra, use \verb|\citetapud| e \verb|\citepapud| (demonstra
627
628 \subsection{M\~ultiplas bibliografias}
629
630 Para separar as refer\~encias (por tipo, cap\~itulo ou tema), use o ambiente \verb|refsection|
631
632 \section{Tipos de entrada bibliogr\~afica}
633
634 Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sin\~oni
635

```

```

636 \subsection{Tipos acadêmicos principais}
637 \begin{itemize}
638   \item Livro / monografia --- \verb|@book| (\verb|@livro|)
639   \item Artigo de periódico --- \verb|@article| (\verb|@artigo|); matéria de jornal --- \verb|@journal|
640   \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
641   \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento --- \verb|@inproceedings|
642   \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo --- \verb|@phdthesis|
643   \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); periódico --- \verb|@article|
644   \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/site --- \verb|@misc|
645 \end{itemize}
646
647 \subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
648 \begin{itemize}
649   \item Patente --- \verb|@patent| (\verb|@patente|)
650   \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato|)
651   \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\verb|@partitura|)
652   \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)
653   \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\verb|@filme|)
654 \end{itemize}
655
656 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
657 \begin{itemize}
658   \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@jogo|)
659   \item Programa de TV / série --- \verb|@tvshow| (\verb|@programadetelevisao|, \verb|@serietv|)
660 \end{itemize}
661
662 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|
663
664 \chapter{Alguns outros exemplo úteis}
665
666 \begin{tcolorbox}[title=Meu Textbox]
667 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
668 \end{tcolorbox}
669
670 \begin{tcolorbox}
671 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem como
672 \end{tcolorbox}
673
674 \begin{figure}[ht]
675   \centering
676   \begin{tikzpicture}
677     \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{image}}
678     \begin{scope}[x={(image.south east)},y={(image.north west)}]
679       % Definindo o textbox dentro da figura
680       \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text align=left]
681       {
682         \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de exemplo]
683           Este textbox fala sobre como inserir um textbox dentro de uma figura usando tikz
684         \end{tcolorbox}
685       };
686     \end{scope}
687   \end{tikzpicture}
688   \caption{Figura com Textbox}
689   \label{fig:figura_com_textbox1}
690   \fonte{Elabora\c{c}\~ao pr\'opria.}

```

```

690 \end{figure}
691
692
693 \begin{figure}[ht]
694     \centering
695     \begin{tcolorbox}
696 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
697 \end{tcolorbox}
698     \caption{Figura com Textbox simples}
699     \label{fig:figura_com_textbox}
700     \fonte{Elabora\c{c}\~ao pr\'opria.}
701 \end{figure}
702
703 \section{Listagens de código e algoritmos}
704
705 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
706
707 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
708 def saudacao(nome):
709     # imprime uma saudação acentuada
710     print(f"Olá, {nome}!")
711
712 saudacao("COPPE")
713 \end{python}
714 \source{Elaboração própria.}
715
716 Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudaca
717
718 \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
719 Olá, COPPE!
720 \end{Verbatim}
721
722 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|, comando
723
724 \begin{algorithm}[H]
725 \caption{Soma dos elementos de um vetor}
726 \label{alg:soma}
727 \Dados{vetor  $v$  com  $n$  elementos}
728 \Resultado{soma  $s$  dos elementos de  $v$ }
729  $s \leftarrow 0$ ;
730 \Para{ $i \leftarrow 1$  \Ate{ }  $n$ }{
731      $s \leftarrow s + v[i]$ ;
732 }
733 \Retorna  $s$ ;
734 \end{algorithm}
735
736 \chapter{Escrevendo matemática}
737
738 A classe carrega \texttt{amsmath} em qualquer motor. Sob \hologo{pdfLaTeX} ela também carr
739
740 \section{Alinhamento (align)}
741 O ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebr
742 \begin{align}
743     f(x) &= (x+1)^2 \\

```

```

744      &= x^2 + 2x + 1 .
745 \end{align}
746
747 \section{Matrizes (matrix)}
748 O \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
749 \begin{equation}
750   A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}
751   \quad
752   \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}
753 \end{equation}
754
755 \section{Subequações (subequations)}
756 Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
757 \begin{subequations}
758 \begin{align}
759   a &= b + c , \\
760   d &= e - f .
761 \end{align}
762 \end{subequations}
763
764 \section{Estruturas alinhadas (array)}
765 O \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
766 \begin{equation}
767   |x| = \left\{ \begin{array}{l}
768     x, \quad \text{se } x \geq 0 , \\
769     -x, \quad \text{caso contrário.}
770   \end{array} \right.
771 \end{equation}
772
773 \section{Operadores (\textbackslash DeclareMathOperator)}
774 Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no pre
775 \begin{verbatim}
776 \usepackage{amsmath}
777 \DeclareMathOperator{\RMSE}{RMSE}
778 \DeclareMathOperator{\diag}{diag}
779 \end{verbatim}
780 Assim, \verb|$\RMSE(y,\hat{y})$| produz  $\RMSE(y,\hat{y})$ , e \verb|$\diag(\lambda_1,\dots
781
782 \section{Unicode e LuaLaTeX (unicode-math)}
783 \label{sec:unicode-math}
784 Ao migrar para \hologo{LuaLaTeX} (ou \hologo{XeLaTeX}), \textbf{não carregue} \texttt{amss
785 \begin{verbatim}
786 \usepackage{amsmath} % a classe já carrega
787 \usepackage{unicode-math} % NÃO use \usepackage{amssymb} junto
788 \setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
789 \end{verbatim}
790 O código clássico continua valendo ( $\mathbb{R}$ ,  $\int_a^b f(x)dx$ ,  $\alpha$ ) e você ai
791
792 \chapter{Referências cruzadas, notas e índices}
793 \label{cap:refs}
794
795 Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,
796
797 \section{Rótulos e referências cruzadas}

```

```

798
799 Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú
800
801 \section{Notas de rodapé}
802
803 Use \verb|\footnote{texto}|\footnote{Como esta nota, que a classe imprime em espaçamento s
804
805 \section{hyperref e marcadores}
806
807 A classe carrega o \verb|hyperref|: o sumário, as citações e os \verb|\ref|/\verb|\autoref|
808
809 \section{URLs}
810
811 Para endereços, use \verb|\url{https://exemplo.org}|: o \verb|hyperref| cria o link e queb
812
813 \section{Índice remissivo}
814
815 Marque os termos com \verb|\index{termo}| e imprima o índice com \verb|\printindex| (carre
816
817 \section{Traduções (NBR 10520)}
818
819 Ao citar um trecho que você mesmo traduziu, coloque a tradução no corpo do texto e escreva
820
821 \chapter{Truques de \LaTeX}
822
823 \section{Comentários mágicos (\texttt{\% !TeX})}
824 Linhas \verb|\% !TeX| no topo do arquivo dizem ao editor (\hologo{TeX}studio, Overleaf, \ho
825 \begin{verbatim}
826 % !TeX program = lualatex
827 % !TeX encoding = UTF-8
828 % !TeX spellcheck = pt_BR
829 % !BIB program = biber
830 \end{verbatim}
831 Com isso o documento compila igual em qualquer ambiente. O \texttt{latexmkrc} cumpre papel
832
833 \section{Espaço engolido}
834 Um comando sem argumento engole o espaço seguinte: \verb|\LaTeX é bom| sai como “\LaTeX é
835
836 \section{Escopo (grupos)}
837 Chaves \verb|\{...}| (ou \verb|\begingroup|\dots\verb|\endgroup|) criam um escopo: mudanças
838
839 \section{Descarregar figuras: \texttt{\textbackslash clearpage} × \texttt{\textbackslash m
840 \verb|\newpage| apenas quebra a página e leva os floats pendentes adiante, acumulando-os.
841
842 \section{\texttt{latexmkrc}, perl e Overleaf}
843 O \verb|latexmk| automatiza a compilação (roda \LaTeX, \verb|biber| e \verb|makeindex| qua
844
845 \chapter{Método Proposto}
846 \chapter{Resultados e Discussões}
847
848 \section{Algumas Demonstrações}
849
850 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alpha
851 \symbl[beta]{Beta}{A palavra Beta more e corrigida}

```

852 \syml[zzbeta]{\beta}{A letra \beta corrigida}
 853 \syml{beta}{A palavra beta}
 854 \syml{alpha}{A palavra alpha}
 855 \syml[alpha]{Alpha}{A palavra Alpha}
 856 \syml[zzalpha]{\alpha}{A letra \alpha corrigida}
 857 \syml[marco]{Marco}{A palavra Marco corrigida}
 858
 859 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exemplos
 860 \abbrev{GoT}{Game of Thrones}
 861 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
 862 \abbrev[iot]{IoT}{IoT ordenado como iot}
 863 \abbrev[IoT]{IoT}{IoT ordenado como IoT}
 864 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
 865 \abbrev{IoT}{IoT com ordenação default}
 866 \abbrev[ITU]{ITU}{ITU mesmo}
 867
 868 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
 869 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
 870
 871 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas automaticamente
 872
 873
 874
 875
 876 \chapter{Conclusões}
 877
 878 \backmatter
 879
 880 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
 881 % partitura --- declarados via sinônimos em português em example.bib.
 882 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
 883 \nocite{jurisprudencia-example}
 884 \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
 885 \nocite{jornalcaderno-example,jornal-example}
 886 \printbibliography
 887
 888
 889
 890 \appendix
 891
 892 \chapter{Um apêndice}
 893
 894 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização de apêndices
 895
 896 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo de complementar
 897
 898
 899 \chapter{Outro apêndice}
 900
 901 \annex
 902
 903
 904 \chapter{Um Anexo}
 905 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização de anexos

```

906
907
908
909 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
910
911 No modelo \CoppeTeX os anexos devem obrigatoriedade vir depois dos apêndices e usam o comand
912
913
914 \chapter{Outro Anexo}
915
916
917 \coppetexfinalpage% colofão institucional (página par, face do contracapa)
918
919 \end{document}
920
921 </example>

```

9 Implementation

9.1 The ‘coppe.cls’ file

```

922 <*class>
923 \def\filename{coppe.dtx}
924 \def\fileversion{v4.0}
925 \def\filedate{2026/05/28}
926 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
927 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
928 % Save the class version and date into private macros immediately, before any
929 % required package can redefine the very common \fileversion / \filedate
930 % global names (LaTeX has no convention reserving them for the host class).
931 % \coppe@version / \coppe@date are what user-facing macros (e.g.
932 % \coppetexfinalpage) should reference.
933 \let\coppe@version\fileversion
934 \let\coppe@date\filedate
935 % Document date stored in private slots so \date can set them without ever
936 % touching TeX's \month/\year primitives (which would break \today). They
937 % default to the current system month/year, so a document without \date
938 % still renders sensibly on the cover and folha de rosto.
939 \edef\coppe@docmonth{\the\month}
940 \edef\coppe@docyear{\the\year}
941 \LoadClass[12pt,a4paper,twoside]{book}
942 % setspace, hyperref and the bibliography engine (biblatex+biber) are loaded at
943 % the END of the class (after every class hook is defined), in that order:
944 % setspace -> hyperref -> biblatex. hyperref must come late so all the macros
945 % it patches (TOC, refs, lists) are already defined, and biblatex must come
946 % after hyperref so its citation links and back-references work.
947 \RequirePackage{hyphenat}
948 \RequirePackage{lastpage}
949 \RequirePackage{ifthen}
950 \RequirePackage{graphicx}
951 \RequirePackage{tabularx}
952 \RequirePackage{booktabs} % regras de tabela profissionais (toprule/midrule/bottomrule)
953 \RequirePackage{longtable} % tabelas que quebram entre páginas
954 \RequirePackage{etoolbox}

```

```

955 \RequirePackage{ltxcmds}
956 \RequirePackage{hologo} % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
957 % Render \TeX, \LaTeX and friends via hologo (\CoppeTeX is a custom logo, kept).
958 \def\TeX{\hologo{TeX}}
959 \def\LaTeX{\hologo{LaTeX}}
960 \def\LaTeXe{\hologo{LaTeXe}}
961 \def\BibTeX{\hologo{BibTeX}}
962 \def\pdfLaTeX{\hologo{pdfLaTeX}}
963 \def\pdfTeX{\hologo{pdfTeX}}
964 \def\LuaLaTeX{\hologo{LuaLaTeX}}
965 \def\XeLaTeX{\hologo{XeLaTeX}}
966 \def\MiKTeX{\hologo{MiKTeX}}
967 % Engine-aware UTF-8 input: pdfLaTeX needs inputenc; LuaLaTeX (and XeTeX)
968 % are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
969 % files may use accented Latin characters (áéíóú) directly. Under pdfLaTeX we also
970 % suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
971 \RequirePackage{iftex}
972 \ifPDFTeX
973   \RequirePackage[utf8]{inputenc}
974   \AtEndOfClass{\ClassWarningNoLine{coppe}{%
975     Although slower, you should try compiling with LuaLaTeX}}
976 \fi
977 \RequirePackage[T1]{fontenc}
978 % Math support. amsmath is universal and works on every engine, so the class
979 % brings it in unconditionally (it must precede hyperref, which is loaded at
980 % the end of the class). amssymb is the legacy AMS symbol package, built
981 % around the 8-bit Type 1 fonts msam/msbm; on a Unicode engine (LuaLaTeX or
982 % XeLaTeX) it clashes with unicode-math, which already provides all of its
983 % symbols through Unicode math codes. So amssymb is loaded ONLY under
984 % pdfLaTeX; Unicode-engine users get a ClassWarning pointing them at
985 % unicode-math (the class does not auto-load unicode-math because choosing
986 % an OpenType math font is a typographic decision that belongs to the author).
987 \RequirePackage{amsmath}
988 \ifPDFTeX
989   \RequirePackage{amssymb}
990 \else
991   % Unicode engine: defer the check to end of document. Only warn if the
992   % user did NOT add either unicode-math (the modern OpenType-math route)
993   % or amssymb (the legacy route, which they would have loaded manually).
994   % This avoids nagging authors who already followed the documented
995   % \usepackage{unicode-math} workflow.
996   \AtEndDocument{%
997     \@ifpackageloaded{unicode-math}{}%
998     \@ifpackageloaded{amssymb}{}%
999     \ClassWarningNoLine{coppe}{%
1000       You are compiling with a Unicode engine (LuaLaTeX or XeLaTeX) and\MessageBreak
1001       neither unicode-math nor amssymb is loaded. AMS-style symbols\MessageBreak
1002       (\string\mathbb, \string\hbar, \string\varnothing, ...) will be undefined.\MessageBreak
1003       Add to your preamble:\MessageBreak
1004       \space\space\string\usepackage{unicode-math}\MessageBreak
1005       \space\space\string\setmathfont{Latin Modern Math}\MessageBreak
1006       Do NOT load amssymb together with unicode-math -- it would clash\MessageBreak
1007       with the OpenType math font setup}}}%
1008 \fi

```

```

1009 % manual2024 margins: 3cm top, 2cm bottom, mirrored 3cm inner / 2cm outer for
1010 % two-sided printing -- realised as inner=outer=2cm plus a 1cm bindingoffset.
1011 % headheight=15pt is given here (not patched later) so geometry's own page
1012 % arithmetic stays self-consistent: fancyhdr needs >= 14.49998pt of header
1013 % height to host the running head, and the standard 12pt default triggers
1014 % its "headheight is too small" warning. Letting geometry know up front
1015 % keeps \topmargin a derived value -- we never touch it by hand.
1016 \RequirePackage[a4paper,twoside,bindingoffset=1cm,headheight=15pt,%
1017 top=3cm,bottom=2cm,inner=2cm,outer=2cm]{geometry}
1018 \RequirePackage{titlesec}
1019 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
1020 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capitulo N").
1021 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
1022 % keep "N Title". \appendix and \annex raise the flag (see below).
1023 \newif\if@coppeappendix \@coppeappendixfalse
1024 \titleformat{\chapter}[hang]
1025   {\normalfont\Large\bfseries}
1026   {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
1027   {1ex}}{\MakeUppercase}
1028 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
1029 \let\coppe@orig@appendix\appendix
1030 \renewcommand\appendix{\coppe@orig@appendix\@coppeappendixtrue}
1031 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
1032 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
1033 % which macro is written (so it is correct at .toc-render time even though the
1034 % appendixname has changed back); the chapter letter is baked in at write time.
1035 \newif\if@coppeannex \@coppeannexfalse
1036 \newcommand\coppe@tocapp[1]{%
1037   \MakeUppercase{\coppestring{appendix}}%
1038   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1039 \newcommand\coppe@tocanx[1]{%
1040   \MakeUppercase{\coppestring{annex}}%
1041   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1042 \def\@chapter[#1]#2{%
1043   \ifnum \c@secnumdepth >\m@ne
1044     \if@mainmatter
1045       \refstepcounter{chapter}%
1046       \typeout{\@chapapp\space\thechapter.}%
1047       \if@coppeappendix
1048         \if@coppeannex
1049           \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
1050         \else
1051           \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
1052         \fi
1053       \else
1054         \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
1055       \fi
1056     \else
1057       \addcontentsline{toc}{chapter}{#1}%
1058     \fi
1059   \else
1060     \addcontentsline{toc}{chapter}{#1}%
1061   \fi
1062   \chaptermark{#1}%

```

```

1063 \addtocontents{lof}{\protect\addvspace{10\p@}}%
1064 \addtocontents{lot}{\protect\addvspace{10\p@}}%
1065 \if@twocolumn
1066 \topnewpage[\@makechapterhead{#2}]%
1067 \else
1068 \makechapterhead{#2}%
1069 \afterheading
1070 \fi}
1071 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
1072 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
1073 \titleformat{name=chapter,numberless}[hang]
1074 {\normalfont\Large\bfseries\filcenter}{0pt}{\MakeUppercase}
1075 \titlespacing{name=chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
1076 % section: ALL CAPS, normal weight.
1077 \titleformat{section}
1078 {\normalfont\large\bfseries}{\thesection}{1ex}{\MakeUppercase}
1079 % subsection: bold (author writes Title Case).
1080 \titleformat{subsection}
1081 {\normalfont\bfseries}{\thesubsection}{1ex}{}
1082 % subsubsection: bold italic.
1083 \titleformat{subsubsection}
1084 {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{}
1085 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
1086 % "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
1087 \RequirePackage{caption}
1088 \captionsetup{font=footnotesize,labelsep=endash}
1089 % subcaption (NÃO subfig/subfigure) for subfigures / subtables; load right after
1090 % caption so it picks up the ABNT captionsetup above.
1091 \RequirePackage{subcaption}
1092 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
1093 % manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
1094 % at the bottom is mandatory). The float package's plaintop style forces the
1095 % caption above the body for figure, table and the new quadro float. -----
1096 \RequirePackage{float}
1097 \floatstyle{plaintop}
1098 \restylefloat{figure}
1099 \restylefloat{table}
1100 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
1101 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
1102 % unlike a tabela (rules on top and bottom only).
1103 \providecommand{\quadroname}{\coppestring{frame}}
1104 \providecommand{\listquadroname}{\coppestring{listframe}}
1105 \newfloat{quadro}{htbp}{loq}[chapter]
1106 \floatname{quadro}{\quadroname}
1107 \providecommand{\quadroautorefname}{\quadroname}
1108 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
1109 % (float's own \l@quadro lays the number/title out incorrectly).
1110 \let\l@quadro\l@figure
1111 \newcommand\listofquadros{%
1112 \chapter*{\listquadroname}%
1113 \addcontentsline{toc}{chapter}{\listquadroname}%
1114 \mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
1115 \starttoc{loq}}
1116 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the

```

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1117 % float as a smaller, single-spaced, centred line; it does NOT advance any
1118 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
1119 % the label is "Fonte" (pt) / "Source" (en).
1120 \newcommand{\cpsourcenam}{\coppestring{source}}
1121 \newcommand{\cpsource}[1]{%
1122   \par\addvspace{\smallskipamount}%
1123   {\footnotesize\singlespacing\centering\cpsourcenam: #1\par}}
1124 \let\cpfonte\cpsource
1125 \@ifundefined{source}{\let\source\cpsource}{}%
1126 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
1127 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
1128 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
1129 \newcommand{\newcoppefloat}[3]{%
1130   {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
1131   \floatname{#1}{#2}%
1132   \expandafter\newcommand\csname listof#1s\endcsname{%
1133     \chapter*{#3}\addcontentsline{toc}{chapter}{#3}%
1134     \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
1135     \@starttoc{lo#1}}}%
1136 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
1137 % (filete) from the left margin; footnotesize is the book default. -----
1138 \let\coppe@orig@footnotetext\@footnotetext
1139 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
1140 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
1141 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
1142 % (manual2024 2.6 / proposta P24). -----
1143 \RequirePackage{enumitem}
1144 \newlist{alineas}{enumerate}{2}
1145 \setlist{alineas,1}{label=\alph*},leftmargin=1.8em,itemsep=0pt}
1146 \setlist{alineas,2}{label=--,leftmargin=1.8em,itemsep=0pt}
1147 % --- Page numbering: arabic in the top OUTER corner on textual pages
1148 % (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
1149 % \frontmatter (unnumbered by default; roman with the numeraisromanos
1150 % class option). -----
1151 \RequirePackage{fancyhdr}
1152 % \headheight is set to 15pt by geometry up above (see the geometry call),
1153 % which is what fancyhdr needs (>= 14.49998pt) for the running head.
1154 \fancypagestyle{coppe}{\fancyhf{}\fancyhead[R0,LE]{\thepage}%
1155   \renewcommand{\headrulewidth}{0pt}}
1156 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
1157 % number may leak onto them (e.g. the verso right after the cover).
1158 \renewcommand{\cleardoublepage}{%
1159   \clearpage
1160   \if@twoside
1161     \ifodd\c@page\else
1162       \hbox{}\thispagestyle{empty}\newpage
1163       \if@twocolumn\hbox{}\newpage\fi
1164     \fi
1165   \fi}
1166 \def\CoppeTeX{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
1167 p\kern-.025em e}}\kern-.08em
1168 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}
1169 \newboolean{maledoc}
1170 \setboolean{maledoc}{false}

```

```

1171 %
1172 % Class options.
1173 % Multilingual model (CoppeTeX 4.x, branch nlinguas). The class holds three
1174 % fixed language slots:
1175 % * main -- the language of the body, captions, and the main abstract.
1176 %          Set by a class option: brazilian (default), english,
1177 %          spanish, french, italian, or any user-supplied language
1178 %          that ships a coppe-lang-<name>.def file.
1179 % * foreign -- the language used by the foreignabstract environment and
1180 %          by \foreign@... registers. Defaults to english (or to
1181 %          brazilian when main=english) for back-compat.
1182 % * third -- optional, loaded on demand with \usecoppelanguage{<name>}
1183 %          in the preamble (for citations or quotations in a third
1184 %          language). Has no class option of its own.
1185 % Note: the legacy \if@english boolean (defined and toggled here in
1186 % v3.x for backward compatibility) was removed in v4.0 -- nothing
1187 % inside the class or in the biblatex style files (.bbx/.cbx/.dbx/.lbr)
1188 % tests it. Code that needs to branch on the main language should test
1189 % \ifx\coppe@mainlang\coppe@englishstr instead.
1190 \def\coppe@mainlang{brazilian}
1191 \def\coppe@foreignlang{english}
1192 \DeclareOption{brazilian}{%
1193   \def\coppe@mainlang{brazilian}\def\coppe@foreignlang{english}}
1194 \DeclareOption{english}{%
1195   \def\coppe@mainlang{english}\def\coppe@foreignlang{brazilian}}
1196 \DeclareOption{spanish}{%
1197   \def\coppe@mainlang{spanish}\def\coppe@foreignlang{english}}
1198 \DeclareOption{french}{%
1199   \def\coppe@mainlang{french}\def\coppe@foreignlang{english}}
1200 \DeclareOption{italian}{%
1201   \def\coppe@mainlang{italian}\def\coppe@foreignlang{english}}
1202 \DeclareOption{msc}{%
1203   \newcommand{\@degree}{M.Sc.}
1204   \newcommand{\@degreename}{Mestrado}
1205   \newcommand{\local@degname}{Mestre}
1206   \newcommand{\foreign@degname}{Master}
1207   \newcommand\local@doctype{Disserta{c c}{\~ a}o}
1208   \newcommand\foreign@doctype{Dissertation}
1209 }
1210 \DeclareOption{dsccexam}{%
1211   \newcommand{\@degree}{D.Sc.}
1212   \newcommand{\@degreename}{Doutorado}
1213   \newcommand{\local@degname}{Doutor}
1214   \newcommand{\foreign@degname}{Doctor}
1215   \setboolean{maledoc}{true}
1216   \newcommand\local@doctype{Exame de Qualifica{c c}{\~ a}o}
1217   \newcommand\foreign@doctype{Qualifying Exam}
1218 }
1219 \DeclareOption{mscexam}{%
1220   \newcommand{\@degree}{M.Sc.}
1221   \newcommand{\@degreename}{Mestrado}
1222   \newcommand{\local@degname}{Mestre}
1223   \newcommand{\foreign@degname}{Master}
1224   \setboolean{maledoc}{true}

```

```

1225 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1226 \newcommand\foreign@doctype{Qualifying Exam}
1227 }
1228 \DeclareOption{dsc}{%
1229 \newcommand{\@degree}{D.Sc.}
1230 \newcommand{\@degreename}{Doutorado}
1231 \newcommand{\local@degname}{Doutor}
1232 \newcommand{\foreign@degname}{Doctor}
1233 \newcommand\local@doctype{Tese}
1234 \newcommand\foreign@doctype{Thesis}
1235 }
1236 \newif\if@coppenumeric\@coppenumericfalse
1237 \DeclareOption{numbers}{\@coppenumerictrue}
1238 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
1239 % pre-textual pages are counted but not numbered (manual2024 2.7).
1240 \newif\if@copperoman\@copperomanfalse
1241 \DeclareOption{numeraisromanos}{\@copperomantrue}

```

Default line spacing is one-and-a-half; the `doublespacing` option switches to double spacing. Since `setspace` is loaded at the end of the class (after `hyperref`), the actual `\onehalfspacing`/`\doublespacing` call is deferred: the option only flips a flag, which the late-load block reads.

```

1242 \newif\if@coppedoublespace\@coppedoublespacefalse
1243 \DeclareOption{doublespacing}{\@coppedoublespacetrue}

1244 \ProcessOptions\relax
1245 % Built-in language names (used by the loader and the dispatcher below).
1246 \def\coppe@brazilianstr{brazilian}
1247 \def\coppe@englishstr{english}
1248 % Load Babel with the foreign language first and the main language last
1249 % (Babel treats the LAST option in the list as the main language). With
1250 % pt-main this produces [english,brazilian]{babel} and with en-main
1251 % [brazilian,english] -- identical to the historical \if@english calls.
1252 \edef\coppe@babelopts{\coppe@foreignlang,\coppe@mainlang}
1253 \expandafter\RequirePackage\expandafter[\coppe@babelopts]{babel}
1254 % --- Multilingual string dispatcher -----
1255 % \copperdefstring{<lang>}{<key>}{<value>} adds one entry to the table for
1256 % language <lang>. The three retrievers below look strings up either at the
1257 % current babel language (\coppestring -- follows \selectlanguage), or pinned
1258 % to the main slot (\coppemainstring) or the foreign slot (\coppeforeignstring).
1259 \providecommand{\copperdefstring}[3]{%
1260 \expandafter\gdef\csname coppe@str@#1@#2\endcsname{#3}}
1261 \providecommand{\coppestring}[1]{\@nameuse{coppe@str@\language@ #1}}
1262 \providecommand{\coppemainstring}[1]{\@nameuse{coppe@str@\coppe@mainlang @#1}}
1263 \providecommand{\coppeforeignstring}[1]{\@nameuse{coppe@str@\coppe@foreignlang @#1}}
1264 % Load an extra language pack at any point in the preamble (third-language
1265 % slot for citations or quotations beyond the main+foreign pair).
1266 \providecommand{\usecoppelanguage}[1]{%
1267 \def\coppe@@cur{#1}%
1268 \ifx\coppe@@cur\coppe@brazilianstr\else
1269 \ifx\coppe@@cur\coppe@englishstr\else
1270 \InputIfFileExists{coppe-lang-#1.def}{%
1271 {\ClassWarning{coppe}{Language pack coppe-lang-#1.def not found}}%
1272 \fi\fi}
1273 % \titlein{<lang>}{<text>}: record the title in any language. Will be

```

```

1274 % consulted by the cover/folha renderers once they are migrated in step 2.
1275 % In step 1 it is just a no-op store -- the historical \title / \foreigntitle
1276 % still drive every visible site, so behavior for pt/en is unchanged.
1277 \providecommand{\titlein}[2]{%
1278   \expandafter\gdef\csmname coppe@title@#1\endcsname{#2}}
1279 % --- Built-in string tables: brazilian + english. The values mirror the
1280 % literals currently scattered through the class; step-1 only populates the
1281 % tables -- no site reads them yet, so the rendered output is unchanged. ----
1282 \copperdefstring{brazilian}{appendix}{\Ap\~endice}
1283 \copperdefstring{english}{appendix}{Appendix}
1284 \copperdefstring{brazilian}{annex}{Anexo}
1285 \copperdefstring{english}{annex}{Annex}
1286 \copperdefstring{brazilian}{frame}{Quadro}
1287 \copperdefstring{english}{frame}{Frame}
1288 \copperdefstring{brazilian}{listframe}{Lista de Quadros}
1289 \copperdefstring{english}{listframe}{List of Frames}
1290 \copperdefstring{brazilian}{source}{Fonte}
1291 \copperdefstring{english}{source}{Source}
1292 \copperdefstring{brazilian}{program}{Programa}
1293 \copperdefstring{english}{program}{Program}
1294 \copperdefstring{brazilian}{listprogram}{Lista de Programas}
1295 \copperdefstring{english}{listprogram}{List of Programs}
1296 \copperdefstring{brazilian}{references}{Refer\~encias}
1297 \copperdefstring{english}{references}{References}
1298 \copperdefstring{brazilian}{in:}{em}
1299 \copperdefstring{english}{in:}{in}
1300 \copperdefstring{brazilian}{art}{Ilustra\c{c}\~ao}
1301 \copperdefstring{english}{art}{Art}
1302 \copperdefstring{brazilian}{platform}{Plataforma}
1303 \copperdefstring{english}{platform}{Platform}
1304 \copperdefstring{brazilian}{directedby}{Dire\c{c}\~ao}
1305 \copperdefstring{english}{directedby}{Directed by}
1306 \copperdefstring{brazilian}{producedby}{Produ\c{c}\~ao}
1307 \copperdefstring{english}{producedby}{Produced by}
1308 \copperdefstring{brazilian}{listabbreviation}{Lista de Abreviaturas e Siglas}
1309 \copperdefstring{english}{listabbreviation}{List of Abbreviations and Acronyms}
1310 \copperdefstring{brazilian}{listsymbol}{Lista de S(\~ i)mbolos}
1311 \copperdefstring{english}{listsymbol}{List of Symbols}
1312 \copperdefstring{brazilian}{glossary}{Gloss(\~ a)rio}
1313 \copperdefstring{english}{glossary}{Glossary}
1314 \copperdefstring{brazilian}{advisor}{Orientador}
1315 \copperdefstring{english}{advisor}{Advisor}
1316 \copperdefstring{brazilian}{advisors}{Orientadores}
1317 \copperdefstring{english}{advisors}{Advisors}
1318 \copperdefstring{brazilian}{dept}{Programa}
1319 \copperdefstring{english}{dept}{Department}
1320 \copperdefstring{brazilian}{approvedmale}{Examinado por}
1321 \copperdefstring{brazilian}{approvedfemale}{Examinada por}
1322 \copperdefstring{english}{approvedmale}{Examined by}
1323 \copperdefstring{english}{approvedfemale}{Examined by}
1324 % Institutional names: always Portuguese on the cover/folha (UFRJ template).
1325 \copperdefstring{brazilian}{universityname}{Universidade Federal do Rio de Janeiro}
1326 \copperdefstring{english}{universityname}{Universidade Federal do Rio de Janeiro}
1327 \copperdefstring{brazilian}{cityname}{Rio de Janeiro}

```

```

1328 \copperdefstring{english} {cityname} {Rio de Janeiro}
1329 \copperdefstring{brazilian}{statename} {RJ}
1330 \copperdefstring{english} {statename} {RJ}
1331 \copperdefstring{brazilian}{countryname} {Brasil}
1332 \copperdefstring{english} {countryname} {Brasil}
1333 % Month names.
1334 \copperdefstring{brazilian}{monthname1} {Janeiro}
1335 \copperdefstring{brazilian}{monthname2} {Fevereiro}
1336 \copperdefstring{brazilian}{monthname3} {Mar{\c c}o}
1337 \copperdefstring{brazilian}{monthname4} {Abril}
1338 \copperdefstring{brazilian}{monthname5} {Maio}
1339 \copperdefstring{brazilian}{monthname6} {Junho}
1340 \copperdefstring{brazilian}{monthname7} {Julho}
1341 \copperdefstring{brazilian}{monthname8} {Agosto}
1342 \copperdefstring{brazilian}{monthname9} {Setembro}
1343 \copperdefstring{brazilian}{monthname10}{Outubro}
1344 \copperdefstring{brazilian}{monthname11}{Novembro}
1345 \copperdefstring{brazilian}{monthname12}{Dezembro}
1346 \copperdefstring{english}{monthname1} {January}
1347 \copperdefstring{english}{monthname2} {February}
1348 \copperdefstring{english}{monthname3} {March}
1349 \copperdefstring{english}{monthname4} {April}
1350 \copperdefstring{english}{monthname5} {May}
1351 \copperdefstring{english}{monthname6} {June}
1352 \copperdefstring{english}{monthname7} {July}
1353 \copperdefstring{english}{monthname8} {August}
1354 \copperdefstring{english}{monthname9} {September}
1355 \copperdefstring{english}{monthname10}{October}
1356 \copperdefstring{english}{monthname11}{November}
1357 \copperdefstring{english}{monthname12}{December}
1358 % algorithm2e package option for this language (empty -> english default).
1359 \copperdefstring{brazilian}{algorithm2eopt}{portuguese}
1360 \copperdefstring{english} {algorithm2eopt}{}
1361 % Babel package name for the foreignabstract \begin{otherlanguage}{...}.
1362 \copperdefstring{brazilian}{babelname}{brazilian}
1363 \copperdefstring{english} {babelname}{english}
1364 % Auto-load language packs for any non-built-in language listed in babelopts.
1365 % Built-ins (brazilian, english) ship their string tables inside coppe.cls
1366 % (defined just above) and need no extra file; others (spanish/french/italian
1367 % /...) ship as coppe-lang-<name>.def alongside the class. The dispatcher
1368 % must already be defined when these run, so the loader sits AFTER the tables.
1369 \@for\coppe@@lang:=\coppe@babelopts\do{%
1370   \edef\coppe@@cur{\coppe@@lang}%
1371   \ifx\coppe@@cur\coppe@@brazilianstr\else
1372     \ifx\coppe@@cur\coppe@@englishstr\else
1373       \InputIfFileExists{coppe-lang-\coppe@@cur.def}
1374       {\ClassInfo{coppe}{Loaded language pack coppe-lang-\coppe@@cur.def}}
1375       {\ClassError{coppe}
1376         {Missing language pack 'coppe-lang-\coppe@@cur.def'.}
1377         {Install it next to coppe.cls, or pick a built-in language.}}%
1378     \fi\fi}
1379 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
1380 % Portuguese alias for short, inline quotations (proposta P3).
1381 \RequirePackage{csquotes}

```

```

1382 \newcommand{\citacao}[1]{\enquote{#1}}
1383 % --- Code listings & algorithms (TODO3 "tratar listagens"): the listings
1384 % package gives the "Programa" listing (caption on top, ABNT R46) with
1385 % \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
1386 % the language styles below are based on the user's listagens.tex. -----
1387 \RequirePackage{xcolor}
1388 \RequirePackage{listings}
1389 \RequirePackage{fancyvrb}
1390 \DeclareFixedFont{\ttb}{T1}{txtt}{bx}{n}{10}
1391 \DeclareFixedFont{\ttm}{T1}{txtt}{m}{n}{10}
1392 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
1393 literate=%
1394 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{e}}1%
1395 {\~}{\~{e}}1 {\~}{\~{e}}1 {\~}{\~{i}}1 {\~}{\~{i}}1 {\~}{\~{o}}1%
1396 {\~}{\~{o}}1 {\~}{\~{o}}1 {\~}{\~{u}}1 {\~}{\~{u}}1 {\~}{\~{c}}1%
1397 {\~}{\~{c}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{E}}1%
1398 {\~}{\~{E}}1 {\~}{\~{I}}1 {\~}{\~{I}}1 {\~}{\~{O}}1 {\~}{\~{O}}1 {\~}{\~{O}}1%
1399 {\~}{\~{U}}1%
1400 }
1401 % Babel-aware listing names, set after listings' own babel setup.
1402 \AtBeginDocument{%
1403 \renewcommand\lstlistingname{\coppestring{program}}%
1404 \renewcommand\lstlistlistingname{\coppestring{listprogram}}%
1405 % \listofprogramas in the coppe list style (heading + ToC + running head).
1406 \newcommand\listofprogramas{%
1407 \chapter*{\lstlistlistingname}%
1408 \addcontentsline{toc}{chapter}{\lstlistlistingname}%
1409 \mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
1410 \starttoc{lol}}
1411 % Per-language styles + environments (python, java, xml, html, prolog).
1412 \newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
1413 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1414 morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1415 emph={MyClass,__init__},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1416 stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1417 extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
1418 escapechar=',postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
1419 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}]%
1420 \csname\@lst @SetFirstNumber\endcsname{\csname\@lst @SaveFirstNumber\endcsname}
1421 \newcommand\pythoninline[1][\pythonstyle\lstinline!#1!]}
1422 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting{#1}{#2}]}
1423 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
1424 numberstyle=\small\ttfamily,numbersep=0.5em,%
1425 morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
1426 keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1427 emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
1428 emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1429 showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
1430 postbreak=\mbox{\textcolor{red}{\hookrightarrow}}\space}}
1431 \lstnewenvironment{xml}[1][\xmlstyle\lstset{#1}]}
1432 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
1433 numberstyle=\small\ttfamily,numbersep=0.5em,%
1434 keywordstyle=\ttb\color{rgb}{0,0,0.5},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1435 stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%

```

```

1436 extendedchars=true,inputencoding=utf8,breaklines=true,%
1437 postbreak=\mbox{\textcolor{red}{\hookrightarrow}\space}}
1438 \lstnewenvironment{html}[1][\{\htmlstyle\lstset{#1}\}]{}
1439 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
1440 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1441 keywordstyle=\ttb\color{rgb}{0,0,0.5},emph={MyClass,__init__},%
1442 emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1443 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1444 breaklines=true,escapechar=',%
1445 postbreak=\mbox{\textcolor{red}{\hookrightarrow}\space}}}
1446 \lstnewenvironment{prolog}[1][\{\prologstyle\lstset{#1}\}]{}
1447 \csname\@lst @SetFirstNumber\endcsname\{\csname\@lst @SaveFirstNumber\endcsname\}
1448 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
1449 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1450 morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1451 emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1452 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1453 breaklines=true,escapechar=',%
1454 postbreak=\mbox{\textcolor{red}{\hookrightarrow}\space}}}
1455 \lstnewenvironment{java}[1][\{\javastyle\lstset{#1}\}]{}
1456 \csname\@lst @SetFirstNumber\endcsname\{\csname\@lst @SaveFirstNumber\endcsname\}
1457 \newcommand\javainline[1][\{\javastyle\lstinline!#1!\}]{}
1458 \newcommand\javaexternal[2][\{\javastyle\lstinputlisting[1]{#2}\}]{}
1459 % fancyvrb's Verbatim environment is available for program output; the
1460 % \DefineVerbatimEnvironment{gxoutput}/{gxoutputs} aliases that used to live
1461 % here were author-personal (gx prefix) and were removed in v4.0. Use, for
1462 % example,
1463 % \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
1464 % ...
1465 % \end{Verbatim}
1466 % (or \DefineVerbatimEnvironment{...}{Verbatim}{...} in the manuscript preamble).
1467 \renewcommand\theFancyVerbLine{\small\ttfamily\arabic{FancyVerbLine}}
1468 % Algorithms: algorithm2e, language driven by the main-slot \coppemainstring
1469 % lookup (key "algorithm2eopt"). Empty means use algorithm2e's default
1470 % language (English) -- needed because algorithm2e errors on an empty option.
1471 \edef\coppe@@algotlang{\coppemainstring{algorithm2eopt}}
1472 \ifx\coppe@@algotlang\empty
1473 \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
1474 \else
1475 \expandafter\RequirePackage\expandafter[\coppe@@algotlang,linesnumbered,lined,algochapter,r
1476 \fi
1477 \SetKwFor{Para}{para}{faça}{fim do para}
1478 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
1479 % setspace + hyperref + biblatex are loaded at the very end of the class
1480 % (see the block just before %</class>), so their hooks see every macro the
1481 % class defines, and the load order setspace -> hyperref -> biblatex is
1482 % respected.

```

`\department` This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```

1483 \newcommand\department[1]{%
1484 \ifthenelse{\equal{#1}{PEB}}{
1485 {\global\def\local@deptname{Engenharia Biom(\ ' e)dica}

```

```

1486 \global\def\foreign@deptname{Biomedical Engineering}}{}
1487 \ifthenelse{\equal{#1}{PEC}}{
1488   {\global\def\local@deptname{Engenharia Civil}
1489     \global\def\foreign@deptname{Civil Engineering}}{}
1490 \ifthenelse{\equal{#1}{PEE}}{
1491   {\global\def\local@deptname{Engenharia El{\'e}trica}
1492     \global\def\foreign@deptname{Electrical Engineering}}{}
1493 \ifthenelse{\equal{#1}{PEM}}{
1494   {\global\def\local@deptname{Engenharia Mec{\'a}nica}
1495     \global\def\foreign@deptname{Mechanical Engineering}}{}
1496 \ifthenelse{\equal{#1}{PEMM}}{
1497   {\global\def\local@deptname{Engenharia Metal{\'urgica e de Materiais}
1498 \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
1499 \ifthenelse{\equal{#1}{PEN}}{
1500   {\global\def\local@deptname{Engenharia Nuclear}
1501     \global\def\foreign@deptname{Nuclear Engineering}}{}
1502 \ifthenelse{\equal{#1}{PEN0}}{
1503   {\global\def\local@deptname{Engenharia Oce{\'a}nica}
1504     \global\def\foreign@deptname{Ocean Engineering}}{}
1505 \ifthenelse{\equal{#1}{PPE}}{
1506   {\global\def\local@deptname{Planejamento Energ{\'e}tico}
1507     \global\def\foreign@deptname{Energy Planning}}{}
1508 \ifthenelse{\equal{#1}{PEP}}{
1509   {\global\def\local@deptname{Engenharia de Produ{\'c}{\~a}{o}}
1510     \global\def\foreign@deptname{Production Engineering}}{}
1511 \ifthenelse{\equal{#1}{PEQ}}{
1512   {\global\def\local@deptname{Engenharia Qu{\'i}mica}
1513     \global\def\foreign@deptname{Chemical Engineering}}{}
1514 \ifthenelse{\equal{#1}{PESC}}{
1515   {\global\def\local@deptname{Engenharia de Sistemas e Computa{\'c}{\~a}{o}}
1516     \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
1517 \ifthenelse{\equal{#1}{PET}}{
1518   {\global\def\local@deptname{Engenharia de Transportes}
1519     \global\def\foreign@deptname{Transportation Engineering}}{}
1520 \ifthenelse{\equal{#1}{PENT}}{
1521   {\global\def\local@deptname{Engenharia de Nanotecnologia}
1522     \global\def\foreign@deptname{Nanotechnology Engineering}}{}
1523 }

```

`\title` Used to enter the title in Brazilian Portuguese. Mirrors into `\coppe@title@brazilian` so that the per-language title slot used by the cover/folha renderers (`\coppe@selecttitle`) picks the right text for a pt-main thesis.

```

1524 \renewcommand\title[1]{%
1525   \global\def\local@title{#1}%
1526   \expandafter\gdef\csname coppe@title@brazilian\endcsname{#1}%
1527 }

```

`\foreigntitle` Used to enter the foreign title. Also mirrors into `\coppe@title@english` so that the cover renders the right text for an english-main thesis.

```

1528 \newcommand\foreigntitle[1]{%
1529   \global\def\foreign@title{#1}%
1530   \expandafter\gdef\csname coppe@title@english\endcsname{#1}%
1531 }

```

`\coppe@selecttitle` Expands to the title stored for the current main language. For pt-main this is `\local@title`; for en-main `\foreign@title`; for any other main language the user must call `\titlein{<lang>}{...}` to populate it.

```
1532 \newcommand\coppe@selecttitle{\@nameuse{coppe@title@\coppe@mainlang}}
```

`\advisor` Defines globally the title, name and academic degree of the advisors.

```
1533 \newcount\@advisor\@advisor0
1534 \newcommand\advisor[4]{%
1535   \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#1}
1536   \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#2}
1537   \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#3}
1538   \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#4}
1539   \global\advance\@advisor by 1
1540   \ifnum\@advisor>1
1541     \renewcommand\local@advisorstring{Orientadores}
1542     \renewcommand\foreign@advisorstring{Advisors}
1543   \fi
1544 }
```

`\examiner`

```
1545 \newcount\@examiner\@examiner0
1546 \newcommand\examiner[3]{%
1547   \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#1\ #2}
1548   \global\advance\@examiner by 1
1549 }
```

`\author` It was redefined to allow the identification of the author's first names and surname.

```
1550 \renewcommand\author[2]{%
1551   \global\def\@authname{#1}
1552   \global\def\@authsurn{#2}
1553 }
```

`\date` Sets the *document's* nominal month and year (used by the cover, the folha de rosto, and the catalog). Stores them in private slots (`\coppe@docmonth` / `\coppe@docyear`) rather than mutating TeX's `\month` / `\year` primitives, which would clobber `\today`.

```
1554 \renewcommand\date[2]{%
1555   \def\coppe@docmonth{#1}%
1556   \def\coppe@docyear{#2}}
```

`\local@monthname`

```
1557 \newcommand\local@monthname{\ifcase\coppe@docmonth\or
1558   Janeiro\or Fevereiro\or Mar{c}o\or Abril\or Maio\or Junho\or
1559   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}
```

`\foreign@monthname`

```
1560 \newcommand\foreign@monthname{\ifcase\coppe@docmonth\or
1561   January\or February\or March\or April\or May\or June\or
1562   July\or August\or September\or October\or November\or December\fi}
```

`\keyword`

```
1563 \newcounter{keywords}
```

```

1564 \newcommand\keyword[1]{%
1565   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
1566   \global\addtocounter{keywords}{1}
1567 }

```

`\freeconfig` This command allows easy changing of core class parameters.

```

1568 \newcommand\freeconfig[6]{%
1569   \providecommand\@degree{}
1570   \renewcommand\@degree{#1}
1571   \providecommand\@degreename{}
1572   \renewcommand\@degreename{#2}
1573   \providecommand\local@degname{}
1574   \renewcommand\local@degname{#3}
1575   \providecommand\foreign@degname{}
1576   \renewcommand\foreign@degname{#4}
1577   \providecommand\local@doctype{}
1578   \renewcommand\local@doctype{#5}
1579   \providecommand\foreign@doctype{}
1580   \renewcommand\foreign@doctype{#6}%
1581 }
1582 %   \end{macrocode}
1583 % \end{macro}
1584 %
1585 % \begin{macro}{\frontmatter}
1586 % The number of pages for both frontmatter and mainmatter printed
1587 % in the cataloging details page is computed by means of simple
1588 % LaTeX\ labels.
1589 %   \begin{macrocode}
1590 \renewcommand\frontmatter{%
1591   \cleardoublepage
1592   \@mainmatterfalse
1593   \pagenumbering{roman}
1594   \thispagestyle{empty}
1595   % folha de rosto = counted page 1 (ABNT: the capa is not counted). An odd
1596   % start keeps the page-counter parity aligned with the physical (recto) side,
1597   % so \cleardoublepage lands every pre-textual element on an odd (front) page.
1598   \setcounter{page}{1}
1599   \makefrontpage
1600   \clearpage
1601   \if@copperoman
1602     \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}%
1603       \renewcommand{\headrulewidth}{0pt}}%
1604     \pagestyle{coppe}%
1605   \else
1606     \fancypagestyle{plain}{\fancyhf{}\renewcommand{\headrulewidth}{0pt}}%
1607     \pagestyle{empty}%
1608   \fi
1609 }

```

`\mainmatter`

```

1610 \renewcommand\mainmatter{%
1611   \coppe@mainBegin
1612   \cleardoublepage
1613   \@mainmattertrue

```

```

1614 \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}%
1615 \renewcommand{\headrulewidth}{0pt}}%
1616 \pagestyle{coppe}%
1617 \pagenumbering{arabic}}

```

\backmatter

```

1618 \renewcommand\backmatter{%
1619 \if@openright
1620 \cleardoublepage
1621 \else
1622 \clearpage
1623 \fi}
1624 %

```

\coppetexfinalpage Optional colophon-style closing page intended to face the rear cover. The macro forces a verso (even) page so that, in a two-sided print, the reader sees the colophon on the left and the inside of the rear cover on the right; pushes the text to the bottom of the page with `\vfill`; and types it in Portuguese unconditionally (the colophon is institutional metadata, not body content, so it is rendered in pt-BR regardless of the thesis main language). Three slot macros are user-overridable: `\coppefinalengine` (engine name; auto-detected via `iftex`), `\coppefinalfont` (text body font; defaults to Computer Modern, the L^AT_EX default), and `\coppefinalmanual` (the reference manual title; defaults to the current COPPE/UFRJ norm). Override them before `\coppetexfinalpage` with `\renewcommand` if the document uses a non-default font or follows an earlier edition of the norm.

```

1625 \providecommand{\coppefinalengine}{%
1626 \ifPDFTeX\hologo{pdfLaTeX}\fi
1627 \ifLuaTeX\hologo{LuaLaTeX}\fi
1628 \ifXeTeX\hologo{XeLaTeX}\fi}
1629 \providecommand{\coppefinalfont}{Computer Modern (pdr~\ao \LaTeX{})}
1630 \providecommand{\coppefinalmanual}{%
1631 \emph{Norma para a Elabora~c~\ao Gr~'afica de Teses e Disserta~c~\oes
1632 da COPPE/UFRJ}}
1633 \newcommand{\coppetexfinalpage}{%
1634 \clearpage
1635 % Force the colophon onto a verso (even) page so it faces the rear cover.
1636 % If the current page is odd (recto), fill it with an empty verso first.
1637 \if@twoside
1638 \ifodd\c@page
1639 \hbox{}\thispagestyle{empty}\newpage
1640 \fi
1641 \fi
1642 \thispagestyle{empty}%
1643 \null\vfill
1644 \begin{otherlanguage}{brazilian}%
1645 \begin{center}
1646 {\footnotesize\setstretch{1.0}}%
1647 \textsc{Colof~\ao}\par
1648 \vspace{0.7em}
1649 \begin{minipage}{0.75\textwidth}\centering
1650 Este documento foi composto em \LaTeX{} com \coppefinalengine{}
1651 em \today, na fonte \coppefinalfont. A bibliografia foi processada

```

```

1652     por Bib\LaTeX{} e Biber. Foi utilizada a classe \CoppeTeX,
1653     vers\~ao\coppe@version, dispon'\i vel no reposit\’orio GitHub
1654     \mbox{\url{https://github.com/COPPE-UFRJ/CoppeTeX}}, em
1655     conformidade com a \coppefinalmanual.
1656     \end{minipage}\par}
1657   \end{center}
1658 \end{otherlanguage}%
1659 \vspace*{2cm}%
1660 \clearpage}

```

`\makecover` The ABNT cover (capa, NBR 14724) suggested in the manual’s Anexo A. It is the institution block — the university, the full COPPE name, and the graduate program taken from `\department` — followed by the author, the title, and the city and year, all centred and uppercased. The optional UFRJ and COPPE logos head the page. The capa is not counted: it precedes the (roman) front matter and is produced before the `titlepage` (folha de rosto).

```

1661 \newcommand\makecover{%
1662   \begin{titlepage}%
1663   \centering
1664   \makebox[\linewidth]{%
1665     \includegraphics[height=2cm]{ufrj-logo}\hfill
1666     \includegraphics[height=2cm]{coppe-logo}}\par
1667   \vspace{1.5cm}
1668   {\singlespacing
1669     \MakeUppercase{\local@universityname}\par
1670     \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de
1671       P{\’ o}s-Gradua{\c c}{\~ a}o e Pesquisa de Engenharia (COPPE)}}\par
1672     \MakeUppercase{Programa de \local@deptname}\par}
1673   \vspace{\stretch{2}}
1674   \textbf{\MakeUppercase{\@authname\ \@authsur}}\par
1675   \vspace{\stretch{2}}
1676   \nohyphens{\MakeUppercase{\coppe@selecttitle}}\par
1677   \vspace{\stretch{3}}
1678   \MakeUppercase{\local@cityname}\par
1679   \coppe@docyear
1680   \end{titlepage}%
1681 }

```

`\maketitle`

```

1682 \renewcommand\maketitle{%
1683   \pagenumbering{alph}
1684   \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
1685   \makecover
1686   \begin{titlepage}
1687   \begin{flushleft}
1688     \vspace*{1.5mm}
1689     \setlength\baselineskip{0pt}
1690     \setlength\parskip{1mm}
1691     \makebox[\linewidth]{%
1692       \includegraphics[height=2cm]{ufrj-logo}\hfill
1693       \includegraphics[height=2cm]{coppe-logo}}
1694     \end{flushleft}
1695     \vspace{1.05cm}

```

```

1696 \begin{center}
1697 \nohyphens{\MakeUppercase{\coppe@selecttitle}}\par
1698 \vspace*{3cm}
1699 \nohyphens{\@authname\ \@authsurname}\par
1700 \end{center}
1701 \vspace*{2.1cm}
1702 \begin{flushright}
1703 \begin{minipage}{8.45cm}
1704 \frontcover@maintext
1705 \end{minipage}\par
1706 \vspace*{7.5mm}
1707 \nohyphens{%
1708 \begin{tabularx}{8.45cm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1709 \local@advisorstring: &
1710 \count1=0
1711 \toks@={}
1712 \@whilenum \count1<\@advisor \do{%
1713 \ifcase\count1 % same as \ifnum0=\count1
1714 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1715 \expandafter\endcsname\expandafter\space%
1716 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1717 \else
1718 \toks@=\expandafter\expandafter\expandafter{%
1719 \expandafter\the\expandafter\toks@%
1720 \expandafter&\expandafter\space%
1721 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1722 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1723 }%
1724 \fi
1725 \advance\count1 by 1}
1726 \the\toks@
1727 \end{tabularx}}\par
1728 \end{flushright}
1729 \vspace*{\fill}
1730 \begin{center}
1731 \local@cityname\par
1732 \local@monthname\ de \coppe@docyear
1733 \end{center}
1734 \end{titlepage}
1735 % Ficha catalográfica on the verso of the folha de rosto (NBR 14724, R57):
1736 % the title page is a recto, so this lands on the immediately following verso,
1737 % ahead of the approval sheet produced by \frontmatter.
1738 \ifthenelse{\boolean{maledoc}}{\thispagestyle{empty}\makecatalog}%
1739 \global\let\maketitle\relax%
1740 \global\let\and\relax}

1741 \newcommand\makefrontpage{%
1742 \begin{center}
1743 \sloppy\nohyphens{\MakeUppercase{\coppe@selecttitle}}\par
1744 \vspace*{7mm}
1745 {\@authname\ \@authsurname}\par
1746 \end{center}\par
1747 \vspace*{4mm}
1748 \frontpage@maintext

```

```

1749 \vspace*{16mm}
1750 \nohyphens{%
1751 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1752 \local@advisorstring: &
1753 \count1=0
1754 \toks@={}
1755 \@whilenum \count1<\@advisor \do{%
1756 \ifcase\count1 % same as \ifnum0=\count1
1757 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1758 \expandafter\endcsname\expandafter\space%
1759 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1760 \else
1761 \toks@=\expandafter\expandafter\expandafter{%
1762 \expandafter\the\expandafter\toks@%
1763 \expandafter&\expandafter\space%
1764 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1765 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1766 }%
1767 \fi
1768 \advance\count1 by 1}
1769 \the\toks@
1770 \end{tabularx}\par
1771 \vspace*{20mm}
1772 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1773 Aprovada por: &
1774 \count1=0
1775 \toks@={}
1776 \@whilenum \count1<\@examiner \do{%
1777 \ifcase\count1 % same as \ifnum0=\count1
1778 \toks@=\expandafter{\csname CoppeExaminer:\the\count1%
1779 \expandafter\endcsname\expandafter\\}
1780 \else
1781 \toks@=\expandafter\expandafter\expandafter{%
1782 \expandafter\the\expandafter\toks@%
1783 \expandafter&\expandafter\space%
1784 \csname CoppeExaminer:\the\count1\expandafter\endcsname%
1785 \expandafter\space\\}
1786 }%
1787 \fi
1788 \advance\count1 by 1}
1789 \the\toks@
1790 \end{tabularx}}\par
1791 \vspace*{\fill}
1792 \frontpage@bottomtext}

1793 \newcommand\coppe@hypersetup{%
1794 \begin{group}
1795 % changes to \toks@ and \count@ are kept local;
1796 % it's not necessary for them, but it is usually the case
1797 % for \count1, because the first ten counters are written
1798 % to the DVI file, thus you got lucky because of PDF output
1799 \toks@={} % in this special case not necessary
1800 \count@=0 %
1801 \@whilenum\count@<\value{keywords}\do{%

```

```

1802 % * a keyword separator is not necessary,
1803 % if there is just one keyword
1804 % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
1805 % at least once, to get rid of the loop depended \count@
1806 \ifcase\count@ % same as \ifnum0=\count@
1807 \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
1808 \else
1809 \toks@=\expandafter\expandafter\expandafter{%
1810 \expandafter\the\expandafter\toks@
1811 \expandafter;\expandafter\space
1812 \csname CoppeKeyword:\the\count@\endcsname
1813 }%
1814 \fi
1815 \advance\count@ by 1 %
1816 }%
1817 \edef\x{\endgroup
1818 \noexpand\hypersetup{%
1819 pdfkeywords={\the\toks@}%
1820 }%
1821 }%
1822 \x
1823 \hypersetup{%
1824 pdfauthor={\@authname\ \@authsur},
1825 pdftitle={\local@title},
1826 pdfsubject={\local@doctype\ de \@degree\ em \local@deptname\ da COPPE/UFRJ},
1827 pdfcreator={LaTeX with CoppeTeX toolkit},
1828 breaklinks={true},
1829 raiselinks={true},
1830 pageanchor={true},
1831 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il;” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain `TeX` command `\isundefined` (cf. the `TeX` FAQ).

```

1832 \newcommand\makecatalog{%
1833 \vspace*{\fill}
1834 \begin{center}
1835 \setlength{\fboxsep}{5mm}
1836 \framebox[120mm][c]{\makebox[5mm][c]{}%
1837 \begin{minipage}[c]{105mm}
1838 \setlength{\parindent}{5mm}
1839 \noindent\sloppy\nohyphens\@authsur,
1840 \nohyphens\@authname\par
1841 \nohyphens{%
1842 \coppe@selecttitle/\@authname\ \@authsur. -- \local@cityname:
1843 UFRJ/COPPE, \coppe@docyear.}\par
1844 \pageref{front:pageno},
1845 \pageref{LastPage}
1846 p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}}{$29,7$cm.\par
1847 % There is an issue here. When the last entry must be split between lines,
1848 % the spacing between it and the next paragraph becomes smaller.
1849 % Should we manually introduce a fixed space? But how could we know that
1850 % a name was split? Is this happening yet?

```

```

1851 \nohyphens{%
1852 \begin{tabularx}{100mm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1853 \local@advisorstring: &
1854 \count1=0
1855 \toks@={}
1856 \@whilenum \count1<\@advisor \do{%
1857 \ifcase\count1 % same as \ifnum0=\count1
1858 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1859 \expandafter\endcsname\expandafter\space%
1860 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1861 \else
1862 \toks@=\expandafter\expandafter\expandafter{%
1863 \expandafter\the\expandafter\toks@
1864 \expandafter&\expandafter\space
1865 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1866 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1867 }%
1868 \fi
1869 \advance\count1 by 1}
1870 \the\toks@
1871 \end{tabularx}}\par
1872 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
1873 UFRJ/COPPE/Programa de \local@deptname, \coppe@docyear.}\par
1874 Refer{\^ e}ncias Bibliogr{\' a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
1875 \count1=0
1876 \count2=1
1877 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
1878 \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
1879 \advance\count1 by 1
1880 \advance\count2 by 1}
1881 I. \csname CoppeAdvisorSurname:0\endcsname,%
1882 \ \csname CoppeAdvisorName:0\endcsname%
1883 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
1884 II. \local@universityname, COPPE, Programa de \local@deptname.
1885 III. T{\' i}tulo.}
1886 \end{minipage}}
1887 \end{center}
1888 \vspace*{\fill}}

```

\dedication

```

1889 \newcommand\dedication[1]{
1890 \gdef\@dedic{#1}
1891 \cleardoublepage
1892 \vspace*{\fill}
1893 \begin{flushright}
1894 \begin{minipage}{60mm}
1895 \raggedleft \it \normalsize \@dedic
1896 \end{minipage}
1897 \end{flushright}}

```

abstract (*env.*) This is a specialization of the abstract in the article standard class.

```

1898 \newenvironment{abstract}{%
1899 \cleardoublepage
1900 \thispagestyle{plain}

```

```

1901 \abstract@toptext\par
1902 \vspace*{8.6mm}
1903 \begin{center}
1904 \sloppy\nohyphens{\MakeUppercase\local@title}\par
1905 \vspace*{13.2mm}
1906 \@authname\ \@authsur \par
1907 \vspace*{7mm}
1908 \local@monthname/\coppe@docyear
1909 \end{center}\par
1910 \vspace*{1.5cm}
1911 \noindent%
1912 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1913 \local@advisorstring: &
1914 \count1=0
1915 \toks@={}
1916 \@whilenum \count1<\@advisor \do{%
1917 \ifcase\count1 % same as \ifnum0=\count1
1918 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1919 \expandafter\endcsname\expandafter\space%
1920 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1921 \else
1922 \toks@=\expandafter\expandafter\expandafter{%
1923 \expandafter\the\expandafter\toks@
1924 \expandafter&\expandafter\space
1925 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1926 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1927 }%
1928 \fi
1929 \advance\count1 by 1}
1930 \the\toks@
1931 \end{tabularx}\par
1932 \vspace*{2mm}
1933 \noindent\local@deptstring: \local@deptname\par
1934 \vspace*{7mm}}{\vspace*{\fill}}

```

foreignabstract (*env.*)

```

1935 \newenvironment{foreignabstract}{%
1936 \cleardoublepage
1937 \thispagestyle{plain}
1938 \begin{otherlanguage}{\coppe@foreignlang}
1939 \foreignabstract@toptext\par
1940 \vspace*{8.6mm}
1941 \begin{center}
1942 \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
1943 \vspace*{13.2mm}
1944 \@authname\ \@authsur \par
1945 \vspace*{7mm}
1946 \foreign@monthname/\coppe@docyear
1947 \end{center}\par
1948 \vspace*{1.5cm}
1949 \noindent%
1950 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1951 \foreign@advisorstring: &
1952 \count1=0

```

```

1953 \toks@={}
1954 \@whilenum \count1<\@advisor \do{%
1955 \ifcase\count1 % same as \ifnum0=\count1
1956 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1957 \expandafter\endcsname\expandafter\space%
1958 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1959 \else
1960 \toks@=\expandafter\expandafter\expandafter{%
1961 \expandafter\the\expandafter\toks@
1962 \expandafter&\expandafter\space
1963 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1964 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1965 }%
1966 \fi
1967 \advance\count1 by 1}
1968 \the\toks@
1969 \end{tabularx}\par
1970 \vspace*{2mm}
1971 \noindent\foreign@deptstring: \foreign@deptname\par
1972 \vspace*{7mm}}{%
1973 \end{otherlanguage}
1974 \vspace*{\fill}
1975 \global\let\@author\@empty
1976 \global\let\@date\@empty
1977 \global\let\foreign@title\@empty
1978 \global\let\foreign@title\relax
1979 \global\let\local@title\@empty
1980 \global\let\local@title\relax
1981 \global\let\author\relax
1982 \global\let\author\relax
1983 \global\let\date\relax}

```

brazilianabstract (*env.*) Optional third abstract for non-pt main languages: a Brazilian Portuguese abstract printed in addition to **abstract** (main language) and **foreignabstract** (English by default). Wraps its body in `\begin{otherlanguage}{brazilian}` so the babel typography (hyphenation, quotation marks) is correct regardless of the current main language. Uses the same `\local@...` Portuguese strings the **abstract** env uses today. Meaningful only when main is neither **brazilian** nor **english**; harmless otherwise (the document just won't include it).

```

1984 \newenvironment{brazilianabstract}{%
1985 \cleardoublepage
1986 \thispagestyle{plain}
1987 \begin{otherlanguage}{brazilian}
1988 \abstract@toptext\par
1989 \vspace*{8.6mm}
1990 \begin{center}
1991 \sloppy\nohyphens{\MakeUppercase\local@title}\par
1992 \vspace*{13.2mm}
1993 \@authname\ \@authsurn \par
1994 \vspace*{7mm}
1995 \local@monthname/\coppe@docyear
1996 \end{center}\par
1997 \vspace*{1.5cm}

```

```

1998 \noindent%
1999 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2000 \local@advisorstring: &
2001 \count1=0
2002 \toks@={}
2003 \@whilenum \count1<\@advisor \do{%
2004 \ifcase\count1
2005 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2006 \expandafter\endcsname\expandafter\space%
2007 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2008 \else
2009 \toks@=\expandafter\expandafter\expandafter{%
2010 \expandafter\the\expandafter\toks@
2011 \expandafter&\expandafter\space
2012 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2013 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2014 }%
2015 \fi
2016 \advance\count1 by 1}
2017 \the\toks@
2018 \end{tabularx}\par
2019 \vspace*{2mm}
2020 \noindent\local@deptstring: \local@deptname\par
2021 \vspace*{7mm}}{%
2022 \end{otherlanguage}
2023 \vspace*{\fill}}

```

\listoffigures

```

2024 \renewcommand\listoffigures{%
2025 \coppe@hasLof
2026 \if@twocolumn
2027 \@restonecoltrue\onecolumn
2028 \else
2029 \@restonecolfalse
2030 \fi
2031 \chapter*{\listfigurename}%
2032 \addcontentsline{toc}{chapter}{\listfigurename}%
2033 \@mkboth{\MakeUppercase\listfigurename}%
2034 {\MakeUppercase\listfigurename}%
2035 \@starttoc{lof}%
2036 \if@restonecol\twocolumn\fi
2037 }

```

\listoftables

```

2038 \renewcommand\listoftables{%
2039 \if@twocolumn
2040 \@restonecoltrue\onecolumn
2041 \else
2042 \@restonecolfalse
2043 \fi
2044 \chapter*{\listtablename}%
2045 \addcontentsline{toc}{chapter}{\listtablename}%
2046 \@mkboth{%
2047 \MakeUppercase\listtablename}%

```

```

2048      {\MakeUppercase\listtablename}%
2049      \@starttoc{lot}%
2050      \if@restonecol\twocolumn\fi
2051      }

```

\printlosymbols

```

2052 \newcommand\printlosymbols{%
2053 \renewcommand\glossaryname{\listsymbolname}%
2054 \@input@{\jobname.los}}

```

\makelosymbols

```

2055 \def\makelosymbols{%
2056   \newwrite\@losfile
2057   \immediate\openout\@losfile=\jobname.syx
2058   \newcommand\syml[3][\@bsphack\begingroup
2059   \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}%
2060       \@sanitize%
2061       \@wrlos{\@tempsymb1}{##2}{##3}\typeout%
2062   {Writing index of symbols file \jobname.syx}%
2063   \let\makelosymbols\@empty%
2064 }%
2065 \@onlypreamble\makelosymbols

2066 \AtBeginDocument{%
2067 \ifpackageloaded{hyperref}{%
2068   \newcommand\@wrlos[3]{%
2069     \protected@write\@losfile{%
2070       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
2071     \endgroup%
2072     \@esphack}}{%
2073   \newcommand\@wrlos[3]{%
2074     \protected@write\@losfile{%
2075       {\string\indexentry{#1[#2] #3}{\thepage}}%
2076     \endgroup%
2077     \@esphack}}}%

```

\printloabbreviations

```

2078 \newcommand\printloabbreviations{%
2079 \renewcommand\glossaryname{\listabbreviationname}%
2080 \@input@{\jobname.lab}}

```

\makeloabbreviations

```

2081 \def\makeloabbreviations{%
2082   \newwrite\@labfile
2083   \immediate\openout\@labfile=\jobname.abx
2084   \newcommand\abbrev[3][\@bsphack\begingroup
2085       \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}
2086       \@sanitize
2087       \@wrlab{\@tempsymb1}{##2}{##3}\typeout
2088   {Writing index of abbreviations file \jobname.abx}%
2089   \let\makeloabbreviations\@empty
2090 }
2091 \@onlypreamble\makeloabbreviations

```

```

2092 \AtBeginDocument{%
2093 \ifpackageloaded{hyperref}{%
2094 \newcommand\wrlab[3]{%
2095 \protected@write\@labfile{%
2096 {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
2097 \endgroup%
2098 \@esphack}}{%
2099 \newcommand\wrlab[3]{%
2100 \protected@write\@labfile{%
2101 {\string\indexentry{#1[#2] #3}{\thepage}}%
2102 \endgroup%
2103 \@esphack}}}%

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
\sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in
the list of abbreviations and acronyms (through \abbrev, so \makeloabbreviations
must be active), then just “SIGLA” on later uses. \sigla*{key} forces the full
form. The first-use flag is set globally so it survives groups (floats, footnotes).

2104 \newcommand\newsigla[3]{%
2105 \@namedef{coppe@sigla@s@#1}{#2}%
2106 \@namedef{coppe@sigla@l@#1}{#3}%
2107 \newcommand\coppe@sigla@reg[1]{%
2108 \@ifundefined{abbrev}{}%
2109 {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}%
2110 \newcommand\coppe@sigla@full[1]{%
2111 \@nameuse{coppe@sigla@l@#1}\space(\@nameuse{coppe@sigla@s@#1})}
2112 \newcommand\coppe@sigla@undef[1]{%
2113 \PackageWarning{coppe}{Siga '#1' nao declarada com \string\newsigla}%
2114 \textbf{???#1???}}
2115 \newcommand\coppe@sigla@first[1]{%
2116 \global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
2117 \newcommand\coppe@sigla@norm[1]{%
2118 \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
2119 {\@ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
2120 {\@nameuse{coppe@sigla@s@#1}}}%
2121 \newcommand\coppe@sigla@star[1]{%
2122 \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
2123 {\global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@full{#1}}}%
2124 \newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}

2125 %%% \AtBeginDocument{%
2126 %%% \ifpackageloaded{hyperref}{%
2127 %%% \def\wrlab#1#2{%
2128 %%% \protected@write\@labfile{%
2129 %%% {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
2130 %%% \endgroup
2131 %%% \@esphack}}{%
2132 %%% \def\wrlab#1#2{%
2133 %%% \protected@write\@labfile{%
2134 %%% {\string\indexentry{[#1] #2}{\arabic{page}}}%
2135 %%% \endgroup
2136 %%% \@esphack}}

2137 % Some macros used to generate cataloging information.
2138 \AtBeginDocument{%

```

```

2139 \ltx@ifpackageloaded{hyperref}{
2140   \def\coppe@bibEnd{%
2141     \immediate\write\@auxout{%
2142       \string\newlabel{bib:end}{\arabic{page}}{\page.\arabic{page}}}%
2143   \def\coppe@bibBegin{%
2144     \immediate\write\@auxout{%
2145       \string\newlabel{bib:begin}{\arabic{page}}{\page.\arabic{page}}}%
2146   \def\coppe@mainBegin{%
2147     \immediate\write\@auxout{%
2148       \string\newlabel{front:pageno}{\Roman{page}}{\page.\roman{page}}}%
2149   \def\coppe@hasLof{%
2150     \immediate\write\@auxout{%
2151       \string\newlabel{cat:lofflag}{\il.}{\page.\roman{page}}}%
2152 }{%
2153   \def\coppe@bibEnd{%
2154     \immediate\write\@auxout{%
2155       \string\newlabel{bib:end}{\arabic{page}}}%
2156   \def\coppe@bibBegin{%
2157     \immediate\write\@auxout{%
2158       \string\newlabel{bib:begin}{\arabic{page}}}%
2159   \def\coppe@mainBegin{%
2160     \immediate\write\@auxout{%
2161       \string\newlabel{front:pageno}{\Roman{page}}}%
2162   \def\coppe@hasLof{%
2163     \immediate\write\@auxout{%
2164       \string\newlabel{cat:lofflag}{\il.}{\page.\roman{page}}}%
2165 }%
2166 }
2167 % The reference list is now produced by biblatex's \printbibliography (see the
2168 % bibliography setup after babel, above). The former thebibliography
2169 % redefinition and its \bibindent are no longer used and have been removed; the
2170 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
2171 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (env.)

```

2172 \newlength{\recuolongquote}%
2173 \setlength{\recuolongquote}{4cm}%
2174 \newenvironment*{longquote}[1][default]{%
2175   \list{}%
2176   \footnotesize%
2177   \addtolength{\leftskip}{\recuolongquote}%
2178   \item[]%
2179   \singlespacing%
2180   \ifthenelse{\not\equal{#1}{default}}{\itshape\selectlanguage{#1}}{}%
2181 }{\endlist}%

2182 \newenvironment{theglossary}{%
2183   \if@twocolumn%
2184     \@restonecoltrue\onecolumn%
2185   \else%
2186     \@restonecolfalse%
2187   \fi%
2188   \@mkboth{\MakeUppercase\glossaryname}%
2189   {\MakeUppercase\glossaryname}%

```

```

2190 \chapter*{\glossaryname}%
2191 \addcontentsline{toc}{chapter}{\glossaryname}
2192 \list{}
2193 {\setlength{\listparindent}{0in}%
2194 \setlength{\labelwidth}{1.0in}%
2195 \setlength{\leftmargin}{1.5in}%
2196 \setlength{\labelsep}{0.5in}%
2197 \setlength{\itemindent}{0in}}%
2198 \sloppy}%
2199 {\if@restonecol\twocolumn\fi%
2200 \endlist}
2201 %
2202 \renewenvironment{theindex}{%
2203 \if@twocolumn
2204 \@restonecolfalse
2205 \else
2206 \@restonecoltrue
2207 \fi
2208 \twocolumn[\@makeschapterhead{\indexname}]%
2209 \mkboth{\MakeUppercase\indexname}%
2210 {\MakeUppercase\indexname}%
2211 \thispagestyle{plain}\parindent\z@
2212 \addcontentsline{toc}{chapter}{\indexname}
2213 \parskip\z@ \@plus .3\p@\relax
2214 \columnseprule \z@
2215 \columnsep 35\p@
2216 \let\item\@idxitem}
2217 {\if@restonecol\onecolumn\else\clearpage\fi}
2218 \newcommand\listabbreviationname{\coppemainstring{listabbreviation}}
2219 \newcommand\listsymbolname{\coppemainstring{listsymbol}}
2220 \newcommand\glossaryname{\coppemainstring{glossary}}
2221 %
2222 \newcommand\local@advisorstring{Orientador}
2223 \newcommand\foreign@advisorstring{Advisor}
2224 \ifthenelse{\boolean{maledoc}}{%
2225 \newcommand\local@approvedname{Examinado por}%
2226 }{%
2227 \newcommand\local@approvedname{Examinada por}%
2228 }
2229 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
2230 \newcommand\local@deptstring{Programa}
2231 \newcommand\foreign@deptstring{Department}
2232 \newcommand\local@cityname{Rio de Janeiro}
2233 \newcommand\local@statename{RJ}
2234 \newcommand\local@countryname{Brasil}
2235 %
2236 \newcommand\frontcover@maintext{
2237 \sloppy\nohyphens{\local@doctype\ de \@degreename\
2238 \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
2239 ao Programa de P{\` o}s-gradua{\c c}{\~ a}o em \local@deptname,
2240 COPPE, da \local@universityname, como parte dos requisitos
2241 necess{\` a}rios {\` a} obten{\c c}{\~ a}o do t{\` i}tulo de
2242 \local@degname\ em \local@deptname.}
2243 }

```

```

2244 %
2245 \newcommand\frontpage@maintext{
2246   \noindent {\MakeUppercase\local@doctype}
2247   \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
2248   \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
2249   DE P{\` O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
2250   UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
2251   NECESS{\` A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
2252   {\MakeUppercase\local@degname} EM CI{\^E}NCIAS EM
2253   {\MakeUppercase\local@deptname.\par}}%
2254 }
2255 %
2256 \newcommand\frontpage@bottomtext{%
2257   \begin{center}
2258     {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
2259     {\MakeUppercase\local@monthname\ DE \coppe@docyear}
2260   \end{center}}%
2261 }
2262 %
2263 \newcommand\abstract@toptext{%
2264   \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
2265   \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
2266   \sloppy\nohyphens{{\` a} COPPE/UFRJ como parte dos requisitos
2267   necess{\` a}rios para a obten{\c c}{\~ a}o do grau de
2268   \local@degname\ em Ci{\^ e}ncias (\@degree)}
2269 }
2270 \newcommand\foreignabstract@toptext{%
2271   \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
2272   COPPE/UFRJ as a partial fulfillment of the requirements for the
2273   degree of \foreign@degname\ of Science (\@degree)}
2274 }
2275 %

```

Late-load packages: setspace, hyperref, biblatex

These three packages *must* be the last things the class loads, and in this exact order:

- **setspace** – supplies `\singlespacing`, `\onehalfspacing` and friends. It is loaded here so the line-spacing state it carries around is settled before **hyperref** patches all the cross-reference and TOC plumbing.
- **hyperref** – patches an enormous number of LaTeX internals (`\section`, `\label`, `\ref`, `\bibcite`, list-of-* commands, ...). Loading it as late as possible is the standard rule of thumb so it sees the final version of those macros, including the ones the **coppe** class redefines (e.g. `\@chapter`, `\listofquadros`).
- **biblatex** – must come after **hyperref** so its citation links and back-references hook into **hyperref** correctly; otherwise the `\cite` commands print as plain (non-clickable) text.

```

2276 \RequirePackage{setspace}
2277 % Apply the default line spacing now that setspace is available. The flag was

```

```

2278 % set above by \DeclareOption{doublespacing}; absent the option, one-and-a-half.
2279 \if@coppedoublespace\doublespacing\else\onehalfspacing\fi
2280 \RequirePackage{hyperref}
2281 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
2282 % former natbib+bibtex setup. The numbers class option selects the numeric
2283 % citation style; otherwise the author-date default is used. natbib=true keeps
2284 % \citet/\citep available for users migrating from natbib.
2285 \if@coppenumeric
2286   \RequirePackage[backend=biber,datamodel=coppe,%
2287     bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
2288 \else
2289   \RequirePackage[backend=biber,datamodel=coppe,%
2290     bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
2291 \fi
2292 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
2293 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
2294 % "Bibliografia" (babel's default \bibname). The title is rendered through
2295 % \coppestring so it follows the current main language in BOTH the chapter
2296 % heading and the .toc entry (biblatex's \bibstring proved unreliable across
2297 % the .toc write, and babel's \bibname caption override was equally fragile).
2298 \defbibheading{bibliography}{%
2299   \chapter*{\coppestring{references}}%
2300   \addcontentsline{toc}{chapter}{\coppestring{references}}}
2301 % The catalog page range (bib:begin/bib:end) was written by the old
2302 % thebibliography redefinition, which biblatex's \printbibliography does not
2303 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
2304 % begin label uses that hook and the end label is emitted by wrapping
2305 % \printbibliography to write it just after the reference list.
2306 \AtBeginBibliography{\coppe@bibBegin}
2307 \let\coppe@orig@printbibliography\printbibliography
2308 \renewcommand\printbibliography[1][]{%
2309   \coppe@orig@printbibliography[#1]%
2310   \coppe@bibEnd}
2311 \</class>
2312 <*glossary>
2313 actual '='
2314 quote '!'
2315 level '>'
2316 %%% delim_0 " , p. "
2317 delim_0 "\\dotfill "
2318 lethead_flag 0
2319 headings_flag 0
2320 preamble
2321 "\n\\begin{theglossary}\n \\makeatletter"
2322 postamble
2323 "\n \\end{theglossary}\n"
2324 </glossary>
2325 <*class>
2326 \newcommand{\annex}{
2327   \renewcommand{\appendixname}{\coppestring{annex}}%
2328   \appendix
2329   \@coppeannextrue
2330 }

```

2331 $\langle$ /class $\rangle$

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```
2332  $\langle$ *dbx $\rangle$ 
2333 \ProvidesFile{coppe.dbx}[2026/05/28 v4.0 CoppeTeX ABNT datamodel]
2334 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
2335 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
2336 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
2337 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
2338 % performance) are already in biblatex's standard datamodel.
2339 \DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
2340 % Custom fields.
2341 \DeclareDatamodelFields[type=field,datatype=literal]{course} % "(... em <course>)"
2342 \DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
2343 \DeclareDatamodelFields[type=field,datatype=literal]{platform} % videogame platform
2344 \DeclareDatamodelFields[type=field,datatype=literal]{scale} % map scale (escala)
2345 \DeclareDatamodelFields[type=list,datatype=literal]{artist} % game/illustration art
2346 \DeclareDatamodelFields[type=list,datatype=literal]{director} % movie/tvshow/video
2347 \DeclareDatamodelFields[type=list,datatype=literal]{producer}
2348 \DeclareDatamodelFields[type=field,datatype=literal]{caderno} % newspaper section
2349 \DeclareDatamodelFields[type=field,datatype=literal]{relator} % jurisprudence rapporteur
2350 \DeclareDatamodelEntryfields{course,coppedegree}
2351 \DeclareDatamodelEntryfields{game,videogame,boardgame,software,movie,tvshow,video}%
2352 {artist,platform,director,producer,version}
2353 \DeclareDatamodelEntryfields[map]{scale}
2354 \DeclareDatamodelEntryfields[article]{caderno}
2355 \DeclareDatamodelEntryfields[jurisdiction]{relator}
2356  $\langle$ /dbx $\rangle$ 
2357  $\langle$ *bbx $\rangle$ 
2358 \ProvidesFile{coppe.bbx}[2026/05/28 v4.0 CoppeTeX ABNT bibliography style]
2359 % The reference list is single-spaced (ABNT 2.4) via \singlespacing in
2360 % \AtBeginBibliography. coppe.cls already loads setspace, but the style must work
2361 % standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
2362 % so load setspace here as well (idempotent).
2363 \RequirePackage{setspace}
2364 % ABNT reference list keeps the date at the end (imprenta), same for author-date
2365 % and numeric modes, so we derive from the standard style, not authoryear.
2366 \RequireBibliographyStyle{standard}
2367
2368 % Note: standard style never dashes repeated authors, which already matches the
2369 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
2370 \ExecuteBibliographyOptions{%
2371 maxbibnames=99, % ABNT 4.3.1.3: list all authors by default
2372 giveninits=false,
2373 uniquename=false,
2374 uniquelist=false,
2375 labeldateparts=true, % needed for author-date extradate (1999a/1999b)
2376 sorting=nyt,
2377 }
2378
2379 % Language strings (Disponível em / Acesso em / In, etc.)
```

```

2380 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
2381 \DeclareLanguageMapping{english}{english-coppe}
2382
2383 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
2384 \DeclareNameAlias{author}{family-given}
2385 \DeclareNameAlias{editor}{family-given}
2386 \DeclareNameAlias{translator}{family-given}
2387 \DeclareNameAlias{default}{family-given}
2388 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
2389 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
2390
2391 % Uppercase the surname (family + prefix) only inside the bibliography list;
2392 % citations keep normal caps (post-2020 rule).
2393 % ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
2394 % between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
2395 % field names; the month stays abbreviated via \mkbibmonth.
2396 \protected\def\coppe@abntdate#1#2#3{%
2397   \iffieldundef{#3}{\stripzeros{\thefield{#3}}\nobreakspace}%
2398   \iffieldundef{#2}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{\space}}%
2399   \iffieldundef{#1}{\stripzeros{\thefield{#1}}}}
2400 \AtBeginBibliography{%
2401   \singlespacing % entries single-spaced internally (2.4)
2402   \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
2403   \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
2404   \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
2405   \let\mkbibdatelong\coppe@abntdate % ABNT date format (drop the "de")
2406   \let\mkbibdateshort\coppe@abntdate
2407 }
2408
2409 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
2410 % (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
2411 % \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
2412 % If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
2413 % article AND the next word are uppercased; otherwise just the first word. The
2414 % delimited macros split off words; \coppeUCkeep strips the trailing-space
2415 % sentinel so no spurious space is left before the following punctuation.
2416 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
2417 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
2418   \coppeUCifart{#1}
2419   {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCtitleB#2\coppeUCstop}}%
2420   {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}%
2421 \protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
2422   \MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}
2423 \protected\def\coppeUCkeep#1 \coppeUCstop{#1}
2424 % \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
2425 \protected\def\coppeUCifart#1{%
2426   \begingroup
2427   \lowercase{\def\coppeUC@w{#1}}%
2428   \edef\coppeUC@w{,\coppeUC@w,%}
2429   \expandafter\in@\expandafter{\coppeUC@w}{,a,o,as,os,um,uma,uns,umas,the,an,}%
2430   \ifin@\aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
2431   \endgroup}
2432 \DeclareFieldFormat{title}{%
2433   \ifbool{test}{\ifnameundef{labelname}} }

```

```

2434     {#1}%
2435     {\mkbibbold{#1}}
2436 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
2437 % format receives the RAW title text (the title format above only sees the
2438 % field-printing commands), so \coppeUCtitle can split off and uppercase the
2439 % first word here.
2440 \DeclareFieldFormat{titlecase}{%
2441   \ifboolexpr{ test {\ifnameundef{labelname}} }
2442     {\coppeUCtitle{#1}}%
2443     {#1}}
2444 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{%
2445 \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
2446 % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
2447 % the title of a whole document, so override those back to bold.
2448 \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
2449 % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
2450 % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
2451 \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
2452 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
2453 \DeclareFieldFormat{booktitle}{\mkbibbold{#1}}
2454 \DeclareFieldFormat{maintitle}{\mkbibbold{#1}}
2455
2456 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
2457 \DeclareFieldFormat{pagetotal}{#1\addnbspace p\adddot}
2458 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\adddot}
2459 \DeclareFieldFormat{pages}{p\adddot\addnbspace#1}
2460
2461 % --- Periodical volume / number prefixes ---
2462 \DeclareFieldFormat[article]{volume}{v\adddot\addnbspace#1}
2463 \DeclareFieldFormat[article]{number}{n\adddot\addnbspace#1}
2464
2465 % --- "In:" before host title (incollection/inproceedings/inbook) ---
2466 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
2467
2468 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
2469 \DeclareFieldFormat[url]{\bibstring{availablefrom}\addcolon\space\url{#1}}
2470
2471 % --- ABNT article driver: no "In:"; Journal, Local, v., n., p., date, DOI/URL.
2472 %   Newspapers (entrysubtype=newspaper, set by the @artigodejornal source map)
2473 %   differ: the newspaper name (journaltitle) and city, then either
2474 %       "<date>. <caderno>, p. <pages>" (when a caderno/section is given), or
2475 %       "p. <pages>, <date>" (page before date, when none),
2476 %   per NBR 6023 §4.2.3.6. ---
2477 \DeclareBibliographyDriver{article}{%
2478   \usebibmacro{bibindex}%
2479   \usebibmacro{begentry}%
2480   \usebibmacro{author/translator+others}%
2481   \setunit{\labelnamepunct}\newblock
2482   \usebibmacro{title}%
2483   \newunit\newblock
2484   \printfield{journaltitle}%
2485   \setunit{\addcomma\space}\printlist{location}%
2486   \iffieldequalstr{entrysubtype}{newspaper}
2487     {\iffieldundef{caderno}

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2488     {\setunit{\addcomma\space}\printfield{pages}%
2489     \setunit{\addcomma\space}\printdate}%
2490     {\setunit{\addcomma\space}\printdate%
2491     \setunit{\adddot\space}\printfield{caderno}%
2492     \setunit{\addcomma\space}\printfield{pages}}}%
2493     {\setunit{\addcomma\space}\printfield{volume}%
2494     \setunit{\addcomma\space}\printfield{number}%
2495     \setunit{\addcomma\space}\printfield{pages}%
2496     \setunit{\addcomma\space}\printdate}%
2497 \newunit\newblock
2498 \printfield{doi}%
2499 \newunit\newblock
2500 \usebibmacro{url+urldate}%
2501 \usebibmacro{finentry}}
2502
2503 % --- incollection: ABNT puts the host organizer before the booktitle, as
2504 % "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
2505 % p. X-Y." The organizer role string is "Org./Orgs." (see the .ltx). ---
2506 \newbibmacro*{coppe:bookeditor}{%
2507   \ifnameundef{editor}
2508   {}
2509   {\printnames{editor}%
2510    \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
2511    \clearname{editor}%
2512    \setunit{\adddot\space}}
2513 \newbibmacro*{coppe:partdriver}{%
2514   \usebibmacro{bibindex}\usebibmacro{begentry}%
2515   \usebibmacro{author/translator+others}%
2516   \setunit{\labelnamepunct}\newblock
2517   \usebibmacro{title}%
2518   \newunit\newblock
2519   \usebibmacro{in:}%
2520   \usebibmacro{coppe:bookeditor}%
2521   \printfield{booktitle}%
2522   \newunit\newblock
2523   \printfield{edition}%
2524   \newunit\newblock
2525   \printlist{location}%
2526   \setunit*{\addcolon\space}\printlist{publisher}%
2527   \setunit*{\addcomma\space}\printdate%
2528   \newunit\newblock
2529   \printfield{pages}%
2530   \setunit{\addcomma\space}\printfield{chapter}%
2531   \newunit\newblock
2532   \usebibmacro{url+urldate}%
2533   \usebibmacro{finentry}}
2534 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
2535 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}
2536
2537 % --- type field: resolve a known bibliography string (babel-aware), else literal ---
2538 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
2539
2540 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
2541 \DeclareFieldFormat{course}{\mkbibparens{\coppestring{in:}\addspace#1}}

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2542 \DeclareBibliographyDriver{thesis}{%
2543   \usebibmacro{bibindex}\usebibmacro{begentry}%
2544   \usebibmacro{author}%
2545   \setunit{\labelnamepunct}\newblock
2546   \usebibmacro{title}%
2547   \newunit\newblock
2548   \printfield{type}\setunit*{\addspace}\printfield{course}%
2549   \newunit\newblock
2550   \printlist{institution}%
2551   \setunit*{\addcomma\space}\printlist{location}%
2552   \setunit*{\addcomma\space}\printdate%
2553   \newunit\newblock
2554   \printfield{pagetotal}%
2555   \newunit\newblock
2556   \printfield{note}%
2557   \newunit\newblock
2558   \usebibmacro{url+urldate}%
2559   \usebibmacro{finentry}}
2560 % --- Games / audiovisual: render the custom fields (artist, platform,
2561 %       director, producer, version) the generic misc driver ignores. ---
2562 \newbibmacro*{coppe:media}{%
2563   \iflistundef{artist}{ }\setunit{\addperiod\space}%
2564   \printtext{\coppestring{art}\addcolon\space}\printlist{artist}}%
2565   \iffieldundef{platform}{ }\setunit{\addperiod\space}%
2566   \printtext{\coppestring{platform}\addcolon\space}\printfield{platform}%
2567   \iflistundef{director}{ }\setunit{\addperiod\space}%
2568   \printtext{\coppestring{directedby}\addcolon\space}\printlist{director}}%
2569   \iflistundef{producer}{ }\setunit{\addperiod\space}%
2570   \printtext{\coppestring{producedby}\addcolon\space}\printlist{producer}}%
2571 \newbibmacro*{coppe:mediadriver}{%
2572   \usebibmacro{bibindex}\usebibmacro{begentry}%
2573   \usebibmacro{author/translator+others}%
2574   \setunit{\labelnamepunct}\newblock
2575   \usebibmacro{title}%
2576   \newunit\newblock
2577   \usebibmacro{coppe:media}%
2578   \newunit\newblock
2579   \printlist{publisher}%
2580   \setunit*{\addcomma\space}\printlist{location}%
2581   \setunit*{\addcomma\space}\printdate%
2582   \newunit\newblock
2583   \printfield{version}%
2584   \newunit\newblock
2585   \printfield{note}%
2586   \newunit\newblock
2587   \usebibmacro{url+urldate}%
2588   \usebibmacro{finentry}}
2589 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriver}}
2590 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriver}}
2591 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriver}}
2592 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriver}}
2593 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriver}}
2594 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
2595 % imprenta) plus the mandatory scale ("Escala 1:N").

```

```

2596 \DeclareFieldFormat{scale}{\bibstring{coppescale}\addnbspace#1}
2597 \DeclareBibliographyDriver{map}{%
2598   \usebibmacro{bibindex}\usebibmacro{begentry}%
2599   \usebibmacro{author/translator+others}%
2600   \setunit{\labelnamepunct}\newblock
2601   \usebibmacro{title}%
2602   \newunit\newblock
2603   \printlist{location}%
2604   \setunit*{\addcolon\space}\printlist{publisher}%
2605   \setunit*{\addcomma\space}\printdate%
2606   \newunit\newblock
2607   \printfield{scale}%
2608   \newunit\newblock
2609   \usebibmacro{url+urldate}%
2610   \usebibmacro{finentry}}
2611 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
2612 % number. date. The holder (depositante) is printed in normal case, not the
2613 % ALL-CAPS family treatment the bibliography gives the leading author.
2614 \newbibmacro*{coppe:depositor}{%
2615   \ifnameundef{holder}
2616   {}
2617   {\printtext{\bibstring{depositor}\addcolon\space}%
2618     \begingroup
2619       \def\mkbibnamefamily##1{##1}%
2620       \def\mkbibnameprefix##1{##1}%
2621       \printnames{holder}%
2622     \endgroup}}
2623 \DeclareBibliographyDriver{patent}{%
2624   \usebibmacro{bibindex}%
2625   \usebibmacro{begentry}%
2626   \usebibmacro{author/translator+others}%
2627   \setunit{\labelnamepunct}\newblock
2628   \usebibmacro{title}%
2629   \newunit\newblock
2630   \usebibmacro{coppe:depositor}%
2631   \newunit\newblock
2632   \printfield{type}%
2633   \setunit*{\addspace}\printfield{number}%
2634   \newunit\newblock
2635   \printdate%
2636   \newunit\newblock
2637   \usebibmacro{url+urldate}%
2638   \usebibmacro{finentry}}
2639 % Exotic ABNT types that biblatex's standard style leaves without a driver:
2640 % give each a baseline (book-like for published docs: norma/partitura; misc for
2641 % the rest) so they print. Per-type ABNT refinements can follow later.
2642 \DeclareBibliographyDriver{standard}{\usedriver{}{book}}
2643 \DeclareBibliographyDriver{music}{\usedriver{}{book}}
2644 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
2645 % the law id + ementa (plain, not bold), then location and date.
2646 \newbibmacro*{coppe:legaldriver}{%
2647   \usebibmacro{bibindex}\usebibmacro{begentry}%
2648   \usebibmacro{author/translator+others}%
2649   \setunit{\labelnamepunct}\newblock

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```

2650 \usebibmacro{title}%
2651 \newunit\newblock
2652 % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
2653 \iffielddundef{relator}{\printtext{Relator\addcolon\space}\printfield{relator}\newunit\ne
2654 \printlist{location}%
2655 \setunit{\addcomma\space}\printdate%
2656 \newunit\newblock
2657 \usebibmacro{url+urldate}%
2658 \usebibmacro{finentry}}
2659 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
2660 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
2661 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
2662 \DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
2663 \DeclareBibliographyDriver{video}{\usedriver{}{misc}}
2664 \DeclareBibliographyDriver{software}{\usedriver{}{misc}}
2665 \DeclareBibliographyDriver{image}{\usedriver{}{misc}}
2666 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
2667 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
2668 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}
2669
2670 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
2671 \DeclareSourcemap{%
2672 \maps[datatype=bibtex]{%
2673 % entry-type PT synonyms -> canonical English
2674 \map{step[typesource=livro,typetarget=book]}
2675 \map{step[typesource=artigo,typetarget=article]}
2676 \map{step[typesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
2677 \map{step[typesource=partedelivro,typetarget=incollection]}
2678 \map{step[typesource=emlivro,typetarget=inbook]}
2679 \map{step[typesource=verbete,typetarget=inreference]}
2680 \map{step[typesource=anais,typetarget=proceedings]}
2681 \map{step[typesource=trabalhodeevento,typetarget=inproceedings]}
2682 \map{step[typesource=periodico,typetarget=periodical]}
2683 \map{step[typesource=relatorio,typetarget=report]}
2684 \map{step[typesource=relatoriotecnico,typetarget=report]}
2685 \map{step[typesource>manual,typetarget>manual]}
2686 \map{step[typesource=correspondencia,typetarget=letter]}
2687 \map{step[typesource=patente,typetarget=patent]}
2688 \map{step[typesource=diverso,typetarget=misc]}
2689 \map{step[typesource=jogo,typetarget=game]}
2690 \map{step[typesource=jogoeletronico,typetarget=videogame]}
2691 \map{step[typesource=jogodetabuleiro,typetarget=boardgame]}
2692 \map{step[typesource=filme,typetarget=movie]}
2693 \map{step[typesource=serietv,typetarget=tvshow]}
2694 \map{step[typesource=programadetelevisao,typetarget=tvshow]}
2695 % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
2696 % except map (declared in coppe.dbx).
2697 \map{step[typesource=legislacao,typetarget=legislation]}
2698 \map{step[typesource=atnormativo,typetarget=legislation]}
2699 \map{step[typesource=jurisprudencia,typetarget=jurisdiction]}
2700 \map{step[typesource=documentojuridico,typetarget=legal]}
2701 \map{step[typesource=documentocivil,typetarget=legal]}
2702 \map{step[typesource=norma,typetarget=standard]}
2703 \map{step[typesource=partitura,typetarget=music]}

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```

2704 \map{\step[typesource=iconografico,typetarget=image]}
2705 \map{\step[typesource=obradearte,typetarget=artwork]}
2706 \map{\step[typesource=tridimensional,typetarget=artwork]}
2707 \map{\step[typesource=cartografico,typetarget=map]}
2708 \map{\step[typesource=mapa,typetarget=map]}
2709 \map{\step[typesource=audiovisual,typetarget=video]}
2710 \map{\step[typesource=conjuntodedados,typetarget=dataset]}
2711 \map{\step[typesource=fasciculo,typetarget=supperperiodical]}
2712 \map{\step[typesource=trabalhoapresentado,typetarget=unpublished]}
2713 % thesis family: retype to thesis and preset the (babel-aware) type word
2714 \map{\step[typesource=tese,typetarget=thesis]}
2715 \map{\step[typesource=dissertacao,typetarget=thesis]}
2716 \map{\step[typesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
2717 \map{\step[typesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
2718 \map{\step[typesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
2719 \map{\step[typesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
2720 \map{\step[typesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
2721 \map{\step[typesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=p
2722 \map{\step[typesource=tesephd,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
2723 \map{\step[typesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
2724 % field PT synonyms (unaccented) -> canonical English
2725 \map{\step[fieldsource=periodico,fieldtarget=journaltitle]}
2726 \map{\step[fieldsource=revista,fieldtarget=journaltitle]}
2727 \map{\step[fieldsource=jornal,fieldtarget=journaltitle]}
2728 \map{\step[fieldsource=titulo,fieldtarget=title]}
2729 \map{\step[fieldsource=subtitulo,fieldtarget=subtitle]}
2730 \map{\step[fieldsource=autor,fieldtarget=author]}
2731 \map{\step[fieldsource=editora,fieldtarget=publisher]}
2732 \map{\step[fieldsource=local,fieldtarget=location]}
2733 \map{\step[fieldsource=ano,fieldtarget=year]}
2734 \map{\step[fieldsource=data,fieldtarget=date]}
2735 \map{\step[fieldsource=edicao,fieldtarget=edition]}
2736 \map{\step[fieldsource=paginas,fieldtarget=pages]}
2737 \map{\step[fieldsource=numero,fieldtarget=number]}
2738 \map{\step[fieldsource=tradutor,fieldtarget=translator]}
2739 \map{\step[fieldsource=organizador,fieldtarget=editor]}
2740 \map{\step[fieldsource=instituicao,fieldtarget=institution]}
2741 \map{\step[fieldsource=tipo,fieldtarget=type]}
2742 \map{\step[fieldsource=nota,fieldtarget=note]}
2743 \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
2744 \map{\step[fieldsource=curso,fieldtarget=course]}
2745 \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
2746 \map{\step[fieldsource=artista,fieldtarget=artist]}
2747 \map{\step[fieldsource=plataforma,fieldtarget=platform]}
2748 \map{\step[fieldsource=diretor,fieldtarget=director]}
2749 \map{\step[fieldsource=produtor,fieldtarget=producer]}
2750 \map{\step[fieldsource=versao,fieldtarget=version]}
2751 \map{\step[fieldsource=escala,fieldtarget=scale]}
2752 }%
2753 }
2754
2755 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
2756 \setlength{\bibhang}{0pt}
2757 \setlength{\bibitemsep}{\baselineskip}

```

```

2758
2759 </bbx>
2760 <*cbx>
2761 \ProvidesFile{coppe.cbx}[2026/05/28 v4.0 CoppeTeX ABNT author-date citations]
2762 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
2763 % author-year cites with normal-caps names (post-2020 rule) out of the box.
2764 \RequireCitationStyle{authoryear-comp}
2765
2766 % --- NBR 10520:2023 author separator -----
2767 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
2768 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
2769 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
2770 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
2771 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
2772 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
2773 \DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
2774 \DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}
2775
2776 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
2777 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
2778 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
2779 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
2780 % so it is never added to the bibliography.
2781 % \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
2782 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]
2783 % \citepapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
2784 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
2785 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
2786 \DeclareRobustCommand{\citetapud}[4][]{%
2787   #2\space(#3)\space\mkbibemph{apud}\space\textcite[#1]{#4}}
2788 \DeclareRobustCommand{\citepapud}[4][]{%
2789   \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
2790     \citeauthor{#4}\addcomma\space\citeyear{#4}%
2791     \ifblank{#1}{ }\addcomma\space\mkpageprefix[pagination]{#1}}}}
2792 </cbx>
2793 <*lbxbr>
2794 \ProvidesFile{brazilian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX brazilian strings]
2795 \InheritBibliographyExtras{brazilian}
2796 \NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale}
2797 \DeclareBibliographyStrings{%
2798   inherit      = {brazilian},
2799   availablefrom = {{Dispon'\{i\}vel em}{Dispon'\{i\}vel em}},
2800   urlseen      = {{Acesso em}{Acesso em}},
2801   in           = {{In}{In}},
2802   editor       = {{Organizador}{Org\adddot}},
2803   editors      = {{Organizadores}{Orgs\adddot}},
2804   depositor    = {{Depositante}{Depositante}},
2805   coppescale   = {{Escala}{Escala}},
2806   mscdiss      = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
2807   dscthesis    = {{Tese de Doutorado}{Tese de Doutorado}},
2808   phdthesis    = {{Tese de Doutorado}{Tese de Doutorado}},
2809   tcc          = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
2810 }
2811 </lbxbr>

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2812 <*lboxen>
2813 \ProvidesFile{english-coppe.lbx}[2026/05/28 v4.0 CoppeTeX english strings]
2814 \InheritBibliographyExtras{english}
2815 \NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale}
2816 \DeclareBibliographyStrings{%
2817   inherit      = {english},
2818   availablefrom = {{Available from}{Available from}},
2819   urlseen      = {{Access on}{Access on}},
2820   in           = {{In}{In}},
2821   depositor    = {{Depositor}{Depositor}},
2822   coppescale   = {{Scale}{Scale}},
2823   mscdiss      = {{M\addot Sc\addot\ Dissertation}{M\addot Sc\addot\ Diss\addot}},
2824   dscthesis    = {{D\addot Sc\addot\ Thesis}{D\addot Sc\addot\ Thesis}},
2825   phdthesis    = {{Ph\addot D\addot\ Thesis}{Ph\addot D\addot\ Thesis}},
2826   tcc          = {{Undergraduate Thesis}{Undergraduate Thesis}},
2827 }
2828 </lboxen>
2829 <*lboxes>
2830 \ProvidesFile{spanish-coppe.lbx}[2026/05/28 v4.0 CoppeTeX spanish strings]
2831 \InheritBibliographyExtras{spanish}
2832 \NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale}
2833 \DeclareBibliographyStrings{%
2834   inherit      = {spanish},
2835   availablefrom = {{Disponible en}{Disponible en}},
2836   urlseen      = {{Consultado el}{Consultado el}},
2837   in           = {{En}{En}},
2838   editor       = {{Editor}{Ed\addot}},
2839   editors      = {{Editores}{Eds\addot}},
2840   depositor    = {{Solicitante}{Solicitante}},
2841   coppescale   = {{Escala}{Escala}},
2842   mscdiss      = {{Tesis de Maestr'\{i\}a}{Tesis de Maestr'\{i\}a}},
2843   dscthesis    = {{Tesis Doctoral}{Tesis Doctoral}},
2844   phdthesis    = {{Tesis Doctoral}{Tesis Doctoral}},
2845   tcc          = {{Trabajo de Fin de Carrera}{Trabajo de Fin de Carrera}},
2846 }
2847 </lboxes>
2848 <*lboxfr>
2849 \ProvidesFile{french-coppe.lbx}[2026/05/28 v4.0 CoppeTeX french strings]
2850 \InheritBibliographyExtras{french}
2851 \NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale}
2852 \DeclareBibliographyStrings{%
2853   inherit      = {french},
2854   availablefrom = {{Disponible \a}{Disponible \a}},
2855   urlseen      = {{Consult\ 'e le}{Consult\ 'e le}},
2856   in           = {{Dans}{Dans}},
2857   editor       = {{\ 'Editeur}{\ 'Ed\addot}},
2858   editors      = {{\ 'Editeurs}{\ 'Eds\addot}},
2859   depositor    = {{D\ 'eposant}{D\ 'eposant}},
2860   coppescale   = {{\ 'Echelle}{\ 'Echelle}},
2861   mscdiss      = {{M\ 'emoire de Master}{M\ 'emoire de Master}},
2862   dscthesis    = {{Th\ 'ese de Doctorat}{Th\ 'ese de Doctorat}},
2863   phdthesis    = {{Th\ 'ese de Doctorat}{Th\ 'ese de Doctorat}},
2864   tcc          = {{M\ 'emoire de fin d\ 'etudes}{M\ 'emoire de fin d\ 'etudes}},
2865 }

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2866 </lbxf>
2867 <*lbxit>
2868 \ProvidesFile{italian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX italian strings]
2869 \InheritBibliographyExtras{italian}
2870 \NewBibliographyString{availablefrom,mscdiss,dscthesis,tcc,depositor,coppescale}
2871 \DeclareBibliographyStrings{%
2872   inherit      = {italian},
2873   availablefrom = {{Disponibile da}{Disponibile da}},
2874   urlseen      = {{Consultato il}{Consultato il}},
2875   in           = {{In}{In}},
2876   editor       = {{Curatore}{Cur\adddot}},
2877   editors      = {{Curatori}{Curr\adddot}},
2878   depositor    = {{Depositante}{Depositante}},
2879   coppescale   = {{Scala}{Scala}},
2880   mscdiss      = {{Tesi di Laurea Magistrale}{Tesi di Laurea Magistrale}},
2881   dscthesis    = {{Tesi di Dottorato}{Tesi di Dottorato}},
2882   phdthesis    = {{Tesi di Dottorato}{Tesi di Dottorato}},
2883   tcc          = {{Tesi di Laurea Triennale}{Tesi di Laurea Triennale}},
2884 }
2885 </lbxit>
2886 <*langes>
2887 %% coppe-lang-spanish.def -- CoppeTeX 4.x Spanish class-string pack.
2888 %% Loaded by coppe.cls when \documentclass[spanish]{coppe} (or any other
2889 %% spanish slot) is active. Populates the dispatcher table for the
2890 %% spanish language and registers the biblatex language mapping.
2891 %% Institutional names (universityname/cityname/statename/countryname)
2892 %% are deliberately kept Portuguese: the COPPE/UFRJ cover and folha de
2893 %% rosto are a Brazilian institutional template that does not translate.
2894 \ProvidesFile{coppe-lang-spanish.def}[2026/05/28 v4.0 CoppeTeX spanish class strings]
2895 \copperdefstring{spanish}{appendix}      {Ap\endice}
2896 \copperdefstring{spanish}{annex}         {Anexo}
2897 \copperdefstring{spanish}{frame}         {Cuadro}
2898 \copperdefstring{spanish}{listframe}     {Lista de Cuadros}
2899 \copperdefstring{spanish}{source}        {Fuente}
2900 \copperdefstring{spanish}{program}       {Programa}
2901 \copperdefstring{spanish}{listprogram}   {Lista de Programas}
2902 \copperdefstring{spanish}{references}    {Referencias}
2903 \copperdefstring{spanish}{in:}          {en}
2904 \copperdefstring{spanish}{art}           {Arte}
2905 \copperdefstring{spanish}{platform}      {Plataforma}
2906 \copperdefstring{spanish}{directedby}    {Direcci'on}
2907 \copperdefstring{spanish}{producedby}    {Producci'on}
2908 \copperdefstring{spanish}{listabbreviation}{Lista de Abreviaturas y Siglas}
2909 \copperdefstring{spanish}{listsymbol}    {Lista de S'imbolos}
2910 \copperdefstring{spanish}{glossary}      {Glosario}
2911 \copperdefstring{spanish}{advisor}       {Director}
2912 \copperdefstring{spanish}{advisors}      {Directores}
2913 \copperdefstring{spanish}{dept}          {Programa}
2914 \copperdefstring{spanish}{approvedmale}  {Examinado por}
2915 \copperdefstring{spanish}{approvedfemale}{Examinada por}
2916 \copperdefstring{spanish}{universityname}{Universidade Federal do Rio de Janeiro}
2917 \copperdefstring{spanish}{cityname}      {Rio de Janeiro}
2918 \copperdefstring{spanish}{statename}     {RJ}
2919 \copperdefstring{spanish}{countryname}   {Brasil}

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2920 \copperdefstring{spanish}{monthname1} {enero}
2921 \copperdefstring{spanish}{monthname2} {febrero}
2922 \copperdefstring{spanish}{monthname3} {marzo}
2923 \copperdefstring{spanish}{monthname4} {abril}
2924 \copperdefstring{spanish}{monthname5} {mayo}
2925 \copperdefstring{spanish}{monthname6} {junio}
2926 \copperdefstring{spanish}{monthname7} {julio}
2927 \copperdefstring{spanish}{monthname8} {agosto}
2928 \copperdefstring{spanish}{monthname9} {septiembre}
2929 \copperdefstring{spanish}{monthname10}{octubre}
2930 \copperdefstring{spanish}{monthname11}{noviembre}
2931 \copperdefstring{spanish}{monthname12}{diciembre}
2932 \copperdefstring{spanish}{algorithm2eopt}{spanish}
2933 \copperdefstring{spanish}{babelname} {spanish}
2934 %% biblatex is loaded by coppe.cls AFTER this .def file is read; defer the
2935 %% language-mapping registration until \begin{document} so biblatex's
2936 %% \DeclareLanguageMapping is available.
2937 \AtBeginDocument{\DeclareLanguageMapping{spanish}{spanish-coppe}}
2938 \langes
2939 \*langfr
2940 %% coppe-lang-french.def -- CoppeTeX 4.x French class-string pack.
2941 %% See coppe-lang-spanish.def for the structure. Institutional names
2942 %% remain Portuguese; only document-content strings translate.
2943 \ProvidesFile{coppe-lang-french.def}[2026/05/28 v4.0 CoppeTeX french class strings]
2944 \copperdefstring{french}{appendix} {Annexe}
2945 \copperdefstring{french}{annex} {Annexe}
2946 \copperdefstring{french}{frame} {Encadr'e}
2947 \copperdefstring{french}{listframe} {Liste des encadr'es}
2948 \copperdefstring{french}{source} {Source}
2949 \copperdefstring{french}{program} {Programme}
2950 \copperdefstring{french}{listprogram} {Liste des programmes}
2951 \copperdefstring{french}{references} {R'ef'erenes}
2952 \copperdefstring{french}{in:} {dans}
2953 \copperdefstring{french}{art} {Illustration}
2954 \copperdefstring{french}{platform} {Plateforme}
2955 \copperdefstring{french}{directedby} {R'ealis'e par}
2956 \copperdefstring{french}{producedby} {Produit par}
2957 \copperdefstring{french}{listabbreviation}{Liste des abr'eviations et acronymes}
2958 \copperdefstring{french}{listsymbol} {Liste des symboles}
2959 \copperdefstring{french}{glossary} {Glossaire}
2960 \copperdefstring{french}{advisor} {Directeur}
2961 \copperdefstring{french}{advisors} {Directeurs}
2962 \copperdefstring{french}{dept} {Programme}
2963 \copperdefstring{french}{approvedmale} {Examin'e par}
2964 \copperdefstring{french}{approvedfemale} {Examin'ee par}
2965 \copperdefstring{french}{universityname}{Universidade Federal do Rio de Janeiro}
2966 \copperdefstring{french}{cityname} {Rio de Janeiro}
2967 \copperdefstring{french}{statename} {RJ}
2968 \copperdefstring{french}{countryname} {Brasil}
2969 \copperdefstring{french}{monthname1} {janvier}
2970 \copperdefstring{french}{monthname2} {f'evrier}
2971 \copperdefstring{french}{monthname3} {mars}
2972 \copperdefstring{french}{monthname4} {avril}
2973 \copperdefstring{french}{monthname5} {mai}

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2974 \copperdefstring{french}{monthname6} {juin}
2975 \copperdefstring{french}{monthname7} {juillet}
2976 \copperdefstring{french}{monthname8} {août}
2977 \copperdefstring{french}{monthname9} {septembre}
2978 \copperdefstring{french}{monthname10}{octobre}
2979 \copperdefstring{french}{monthname11}{novembre}
2980 \copperdefstring{french}{monthname12}{décembre}
2981 \copperdefstring{french}{algorithm2eopt}{french}
2982 \copperdefstring{french}{babelname} {french}
2983 \AtBeginDocument{\DeclareLanguageMapping{french}{french-coppe}}
2984 \end{langfr}
2985 \end{langit}
2986 %% coppe-lang-italian.def -- CoppeTeX 4.x Italian class-string pack.
2987 %% See coppe-lang-spanish.def for the structure. The algorithm2e
2988 %% language option for Italian is spelled "italiano" -- that's the
2989 %% spelling algorithm2e ships, even though babel calls the language
2990 %% "italian".
2991 \ProvidesFile{coppe-lang-italian.def}[2026/05/28 v4.0 CoppeTeX italian class strings]
2992 \copperdefstring{italian}{appendix} {Appendice}
2993 \copperdefstring{italian}{annex} {Allegato}
2994 \copperdefstring{italian}{frame} {Riquadro}
2995 \copperdefstring{italian}{listframe} {Elenco dei riquadri}
2996 \copperdefstring{italian}{source} {Fonte}
2997 \copperdefstring{italian}{program} {Programma}
2998 \copperdefstring{italian}{listprogram} {Elenco dei programmi}
2999 \copperdefstring{italian}{references} {Riferimenti}
3000 \copperdefstring{italian}{in:} {in}
3001 \copperdefstring{italian}{art} {Illustrazione}
3002 \copperdefstring{italian}{platform} {Piattaforma}
3003 \copperdefstring{italian}{directedby} {Regia di}
3004 \copperdefstring{italian}{producedby} {Prodotto da}
3005 \copperdefstring{italian}{listabbreviation} {Elenco di abbreviazioni e sigle}
3006 \copperdefstring{italian}{listsymbol} {Elenco dei simboli}
3007 \copperdefstring{italian}{glossary} {Glossario}
3008 \copperdefstring{italian}{advisor} {Relatore}
3009 \copperdefstring{italian}{advisors} {Relatori}
3010 \copperdefstring{italian}{dept} {Programma}
3011 \copperdefstring{italian}{approvedmale} {Esaminato da}
3012 \copperdefstring{italian}{approvedfemale} {Esaminata da}
3013 \copperdefstring{italian}{universityname} {Universidade Federal do Rio de Janeiro}
3014 \copperdefstring{italian}{cityname} {Rio de Janeiro}
3015 \copperdefstring{italian}{statename} {RJ}
3016 \copperdefstring{italian}{countryname} {Brasil}
3017 \copperdefstring{italian}{monthname1} {gennaio}
3018 \copperdefstring{italian}{monthname2} {febbraio}
3019 \copperdefstring{italian}{monthname3} {marzo}
3020 \copperdefstring{italian}{monthname4} {aprile}
3021 \copperdefstring{italian}{monthname5} {maggio}
3022 \copperdefstring{italian}{monthname6} {giugno}
3023 \copperdefstring{italian}{monthname7} {luglio}
3024 \copperdefstring{italian}{monthname8} {agosto}
3025 \copperdefstring{italian}{monthname9} {settembre}
3026 \copperdefstring{italian}{monthname10}{ottobre}
3027 \copperdefstring{italian}{monthname11}{novembre}

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3028 \copperdefstring{italian}{monthname12}{dicembre}
3029 \copperdefstring{italian}{algorithm2eopt}{italiano}
3030 \copperdefstring{italian}{babelname} {italian}
3031 \AtBeginDocument{\DeclareLanguageMapping{italian}{italian-coppe}}
3032 \langit)
3033 <*numbbx>
3034 \ProvidesFile{coppe-numeric.bbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric bibliography]
3035 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
3036 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
3037 % the PT->EN source map and the language mappings) and only adds the [n] list
3038 % label and the numeric sorting on top.
3039 \RequireBibliographyStyle{coppe}
3040
3041 % ABNT numeric system lists references in the order they are first cited
3042 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
3043 % numbered list. 'labelnumber' generates the [n] field.
3044 \ExecuteBibliographyOptions{labelnumber,sorting=none}
3045
3046 % Bracketed labels, both in the list and for any shorthand entries.
3047 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
3048 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
3049
3050 % Reference list driven by the [n] label (copied from biblatex's numeric.bbx),
3051 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
3052 \defbibenvironment{bibliography}
3053 {
3054   \list
3055     {\printtext[labelnumberwidth]{%
3056       \printfield{labelprefix}%
3057       \printfield{labelnumber}}}
3058     {\setlength{\labelwidth}{\labelnumberwidth}%
3059      \setlength{\leftmargin}{\labelwidth}%
3060      \setlength{\labelsep}{\biblabelsep}%
3061      \addtolength{\leftmargin}{\labelsep}%
3062      \setlength{\itemsep}{\bibitemsep}%
3063      \setlength{\parsep}{\bibparsep}}%
3064     {\renewcommand*{\makelabel}[1]{\hss#1}}
3065   {\endlist}
3066   {\item}
3067 }
3068 <*numcbx>
3069 \ProvidesFile{coppe-numeric.cbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric citations]
3070 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
3071 % numbers [1], compressed and sorted within a single cite group. This is the
3072 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
3073 % bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and
3074 % \citet -> Author [1].
3075 \RequireCitationStyle{numeric-comp}
3076 </numcbx>

```

Acknowledgments

Thanks to all COPPE_{TE}X users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their helpful comments. The authors would like to thank the National Council for Scientific and Technological Development (CNPq) of Brazil.

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Change History

v0.0	General: Creation Date.	1	v1.0	General: First COPPET _{EX} release. .	1
v0.1	General: Documentation: bibliography fixed, title translation.	1	v2.0	General: COPPET _{EX} release 2.0. . .	1
	Sourceforge submission.	1	v2.1	General: COPPET _{EX} release 2.1: Matching the new rules.	1
v0.2	General: Unification of the code for the list of symbols and abbreviations.	1		\advisor: Advisors, co-advisors, co-co-advisors, etc., all of them are simply considered advisors.	37
v0.3	General: Added ‘draft’ option. . . .	1	v2.2	General: Matching new guidelines, including new logo.	1
	\maketitle: Added number of examiners test.	41	v2.2.2	General: Fixed some text constants in .bib and documented it here.	1
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v0.4	General: Beta documentation. . . .	1		New approval page layout. . . .	42
v0.5	abstract: Changed from macro to environment.	45		\department: Added new course on Nanotechnology.	37
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	foreignabstract: Changed from macro to environment.	46	v3.1	\department: Included a sort key	

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v3.2	General: Fixed version problem			General: Class-wide string
	between cls and dtx.	1		dispatcher (copperdefstring,
v3.3	General: Extend abbrev to work like			coppestring, coppemainstring,
	symb and accept a sorting key . .	1		coppeforeignstring) replaces
v3.4	General: Some examples for			scattered iflanguage and if
	figures, tables, longtables, etc. .	1		english branches; pt/en
				outputs remain byte-identical. .
v3.4.1	General: Fix english option to			1
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v3.5	General: New command annex . .	66		options (brazilian, english,
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